CHAOS

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Part One

INTRODUCTION

Chapter 1
WHAT IS CHAOS?

There is no generally accepted definition of chaos. We begin with a few historical remarks.

In 1963 E. N. Lorenz published his numerical observations on a simplified model of thermal convection[1], a model which now carries his name. He discovered that in this completely deterministic system of three ordinary differential equations, all nonperiodic solutions were. bounded but. unstahle, i.e. they underwent irregular fluctuations without any element of randomness introduced from the outside. In 197 1 D. Ruelle and F. **Takens**[2] coined, the term "strange attractor" for dissipative dynamical systems, though they were unaware of the Lorenz model as the first example having strange attractors. They further suggested a new mechanism for the onset of turbulence, **Li and Yorke**[3] seemed to be the first to introduce the word "chaos" into the mathematical literature to, denote the apparently 'random output of some mappings, although the use of "chaos" in physics dates back to L. Boltzmann in another context, not unrelated to its present usage. In an excellent review published in **1976**[4], R. May called attention to the very complicated dynamics including period-doubling and chaos in some very simple population models. Then came the discovery by **Feigenbaum**[5] of the scaling properties and universal constants in one-dimensional mappings, as well, as the introduction of renor malization group idea into this field by him. The work of Feigenbaum triggered an upsurge of research interest among physicists, as witnessed by the bibliography in this volume.

There has been another course of events leading to the domain of chaos, namely, the study of nonintegrable Hamiltonian systems in classical mechanics. By the end of the last century the development of celestial mechanics and the foundation of statistical mechanics had posed a number o f deep problems in classical dynamics. Since the success of relativity and quantum theory as well as the rapid progress of modern technology absorbed the attention of almost all physicists, these difficult

problems were left to mathematicians for calm study for more than half a century. Their efforts crystallized in the formulation of the so-called KAM theorem on nearly integrable Hamiltonian systems in the early 60's (see Chapter 2). Numerical studies on what happens when the conditions of the KAM theorem fail have revealed an abundance of random motion in nonintegrable systems[6].

The phenomena related to the occurrence of randomness and unpredictability in completely deterministic systems have been called "dynamical stochasticity", "deterministic chaos", "selfgenerated noise", "lntrinslc stochasticity", "Hamiltonian stochasticity" and so on by various authors. We prefer the word "chaos" as the shortest and use it mainly in the context of dissipative systems.

Actually there are two distinct classes of deterministic equations, and at least two different levels of stochasticity in physics.

On the fundamental "microscopic" level we have the law of dynamics, represented by Newton's equations in classical mechanics. In the absence of external timedependent forces, they describe reversible movement of conservative systems. Does stochasticity and the necessity of a statistical description occur spontaneously, when the system gets complicated enough? This is the old problem of the relation between mechanics and statistical mechanics. In the hands of mathematicians, as part of ergodlc theory, dynamical system theory and the qualitative theory of differential equations, significant progress has been reached in understanding this problem. We shall touch briefly on the essentials in Chapter 2, because in studying dissipative systems many concepts are borrowed from the well-shaped mathematics of stochasticity in Hamiltonlan systems.

On the other hand, there are many macroscopic equations in physics, the first example being the Navier-Stokes equations describing the velocity field $\mathbf{V}$ of a fluid :

$$\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V}\nabla)\mathbf{V} = -\frac{1}{\rho}\nabla p + \nu\nabla^2 \mathbf{V} \tag{1-1}$$

where p, ν and ρ denote respectively pressure, kinematic viscosity and density. For another example one may take the reactiondiffusion equations describing chemical reactions in an inhomogeneous medium:

$$\frac{\partial x_i}{\partial t} = f_i(x_1, \ldots, x_n) + D_i\nabla^2 x_i \quad , \qquad i = 1, \ldots, n \tag{1-2}$$

where the reaction kinetics is represented by functions f_i and the diffusion caused by spatial inhomogeneity is described by the second term, D_i being the diffusion constant of the i-th component x_i.

These nonlinear (due to the $(\mathbf{V}\nabla)\mathbf{V}$ term or the functions f_i) and dissipative (due to the v or D_i terms) evolution equations are irreversible in nature. Although they can be derived, in principle, from microscopic dynamics by making statistical assumptions or coarse-graining at certain steps of the derivation, the equations themselves, supplemented wlth appropriate boundary and initial conditions, are apparently deterministic. It is a basic experimental fact that under certain conditions a fluid or a reacting system may undergo transitions into states of more and more erratic motion, which in turn necessitates a statistical description. This leads to another long-standing problem in physics: the problem of turbulence. Does the onset of turbulence follow from such macroscopic deterministic equations as (l-l) or (1-2)? And furthermore, can one characterize the state of developed turbulence startlng from these same equations? To a certain extent the recent upsurge of interest in chaos was roused by the hope to understand these questions. We shall devote Chapter 3 to the problem of turbulence.

While equations (l-1) and (l-2) are partial differential equations, i.e. systems with infiite degrees of freedom, most models studied so far are low-dimensional, e.g., one- or two-dimensional mappings, ordinary differential equations with three or more variables, etc. A moral drawn from recent studies on chaos in low-dimensional systems consists of the belief that at least the onset of turbulent behaviour in dissipative systems, no matter how large the original phase space, may be described by motions in subspaces of much lower dimension; called attractors. In other words, it is dissipation that realizes the contraction of description in a natural way: a vast number of modes die out due to dissipation, only those-spanning the attractors need be taken into account in modelling the system. This kind of simplification cannot take place in Hamiltonian systems owing to the preservation of phase volume (Liouville's theorem). Moreover, dissipation is unavoidable in most experiments. This explains why chaos in dissipative systems has attracted the attention of more and more physicists.

To conclude this introductory chapter we emphasize that chaos is not to be equated simply with disorder. It is more appropriate to consider chaos as a kind of order without periodicity. Within generally chaotic regimes one can discover patterns of ordered motion interspersed with chaos at smaller males, provided sufficiently high resolving power is reached in numerical or laboratory experiments. Instead of the usual spatial or temporal periodicity, there appears some kind of scale invariance which opens the possibility for renormalizationgroup considerations in studying chaotic transitions.

References and guide to the bibliogmphy

1. Lorenz (1963), Paper 14 in this volume, see also book B32 (Sparrow, 1982).

2. Ruelle and Takens (1971), Paper 5 in this volume.
3. Li and Yorke (1975), Paper 10 in this volume.
4. May (1976), Paper 6 in this volume. This is a good starting point for a first acquaintance with chaos.
5. Feigenbaum (1978, 1979), Papers 7 and 8 in this volume.
6. See, e.g., Walker and Ford (1969). Paper 2 in this volume, and references given at the end of Chapter 2.
7. For semi-popular articles on chaos see, eg., Ruelle (1980b), Feigenbaum (1980), GBL (1981), Hofstadter (1981). and Kadanoff (1983b).
8. Among recent books on chaos we draw attention to B31 (Lichtenberg and Lieberman, 1982) and B36 (Guckenhehner and Holmes, (1983), both from the Springer Applied Mathematics Series. B31 deals basically with conservative systems, and is more readable for physicists, while B36 distinguishes itself in mathematical clarity.
9. To our knowledge no book on the physics of chaotic phenomenom has appeared yet, but there are many reviews in physics journals, e.g., Eckmann (1981), Ott (1981), Shaw (1981), Hu (1982a), Tomita (1982a), Hao (1983), Huberman (1983a), and especially, Swhmey (1983) for a review on experimental observations of chaos. One may also consult the proceedings of many conferences listed in the first part of the bibliography.
10. Since we shall not touch at all the problem of quantum chaos in this volume, we mention. here the book B4 (Casati and Ford, 1977) and,, two reviews: Chiriiov et al,, (1981), Zaslavsky (1981), as well as a few recent papers: Berman and Zaslavsky (1977, 1982), Shepelyansky (1981, 1983), Bellissard ***et al,*** (1982), Fishman ***et al,*** (1982), Hogg and Huberman (1982, 1983), Jaffe and Reinhardt (1982), Shapiro and Child (1982), Weissman and Jortner (1982), Berman and Kolovsky (1983), Casati and Guarneri (1983), and Graham (1983b).

Chapter 2
KAM THEOREM AND STOCHASTICITY IN CLASSICAL HAMILTONIAN SYSTEMS

As mentioned before, stochasticity in Hamiltonian systems has become a well-shaped chapter of mathematics and there are many excellent reviews and books on this subject! . This chapter is included since some notions formulated for Hamiltonian systems remain useful for 'understanding chaotic behaviour in dissipative systems.

To begin with we recall a few concepts from analytic mechanics. The motion of a classical conservative system of N degrees of freedom with Hamiltonian function

$$H = H(p_1, \ldots, p_N; q_1, \ldots, q_N) \tag{2-1}$$

is described by the Hamilton's canonical equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad i = 1, \ldots, N \tag{2-2}$$

If there exist successive canonical transformations changing $\{p_i, q_i\}$ into a new set of canonical variables $\{J_i, Q_i\}$ such that in terms of these new variables the Hamiltonian function depends only on the J_i's, but not on the Q_i's, i.e., all the Q_i's become cyclic variables:

$$H = H(J_1, \ldots, J_N), \tag{2-3}$$

then the corresponding canonical equations

$$\dot{Q}_i = \frac{\partial H}{\partial J_i} = \Omega_i(J_1, \ldots, J_N)$$

$$\dot{J}_i = \frac{\partial H}{\partial Q_i} = 0 \tag{2-i}$$

can readily be integrated to glve

$$Q_i(t) = \Omega_i t + Q_i(0) \quad ,$$
$$J_i(t) = J_i(0) \quad . \qquad (2\text{-}5)$$

Now going back to the old variables, one gets 2N combinations of $\{p_i, q_i\}$ and t:

$$Q_i(0) = Q_i(p_1(t), \ldots, p_N(t); q_1(t), \ldots, q_N(t); t)$$
$$J_i(0) = J_i(p_1(t), \ldots, p_N(t); q_1(t), \ldots, q_N(t)) \qquad (2\text{-}6)$$

which do not change with time. In other words, we have solved the equations of motion completely and obtained 2N constants (or integrals, or invariants) of the motion. Such Hamiltonian systems are said to be *integrable*. In fact, the existence of N independent integrals of motion suffices to make the system integrable.

A qualltative picture of the motion of an integrable system looks very much like that of a system of coupled oscillators, as can be seen from (2-S). To keep the motion in a finite region of the phase space, the linearly growing $Q_i(t)$'s must appear as arguments of periodic functions. When N = 1 the motion can be visualized as rotation around a circle of'radlus $\sqrt{2J(0)}$ with constant angular velocity Ω; therefore the bounded motion of a system with one degree of freedom isalwaysperiodic. When N = 2, there are two radii determined by $J_1(0)$ and $J_2(0)$ with angular velocities Ω_1 and Ω_2, so the motion is confined to a two-dimensional toroidal surface or a 2-torus. Resides periodic motion a new possibility appears. When the ratio Ω_1 / Ω_2 happens to be an irrational number, the motion can no longer be periodic: the trajectory winds up the 2-torus densely and endlessly. This kind of motion is called *quasiperiodic* (or conditionally periodic in the Russian literature). In general, the motion of an integrable Hamiltonian system with N degrees of freedom is quasiperiodic and is confined to an N-torus. Therefore, integrable systems have nothing to do with the requirements of statistical mechanics, since the dimension of the constant energy surface (2N – 1) is larger than that of the torus whenever N > 1, and the trajectory can in no way fill up the energy surface, not to mention the equal probability assumption of microcanonical ensemble.

Two questions arise immediately:

1. Are there many integrable systems among all Hamiltonian systems?

2. What happens with the qualitative picture of motion when the system is made slightly nonintegrable, i.e., when the Hamiltonian becomes

$$H = H_0 + V \quad , \qquad (2\text{-}7)$$

where H_0 is integrable and V contains a small parameter[2]?

The answer to the first question is definitely negatives. It was found that integrability is an exceptional property for Hamiltonian systems whenever the number of degrees of freedom gets largerthan two. Integrable systems are so rare that in general it is impossible toapproximate a nonintegrable Hamiltonian system by a series of integrable ones. This statement is to be compared with irrational numbers which can always be approached from both sides by sequencesof rationals, because rational numbers are dense on the number axis, though having zero measures.

The answer to the second question is provided by the KAM **theorem**[3], first enunciated by Kolmogorov in 1954 and completely proved by Arnold and Moser in the early 60's. The proof requited a successful treatment of the small divisor problem in perturbative solutions to the classical many-body problem. The mathematical prerequisite goes beyond that of an average physicist, so we confine ourselves to a loose formulation of the theorem and then turn to its physical implications.

KAM proved that provided the following two conditions hold:

(a) The perturbation V causing nonintegrabillty in (2-7) is small (we ignore the precise formulation for smallness);

(b) The frequencies Ω_i of the unperturbed integrable Hamiltonian satisfy the noncorrelated or nonresonance condition

$$\frac{\partial(\Omega_1, \ldots, \Omega_N)}{\partial(J_1, \ldots, J_N)} = 0 \quad , \qquad (2\text{-}8)$$

then the motion is still confined to an N-torus except for a negligible set (of measure zero) of initial conditions which may lead to wandering motion on the energy surface. These N-tori, now called *KAM surfaces*, or *KAM curves* if seen in plane sections, may be slightly distorted compared to-that of the $V = 0$ case; nevertheless the qualitative picture of the motion remains much the same as the unperturbed integrable system.

If we follow the exceptional trajectories mentioned 'in the KAM theorem, then a qualitatively new phenomenon appears for systems with N degrees of freedom, $N > 2$. This follows from the fact that the boundary of the constant energy surface must be of dimension $2N - 2$ and when N < 2N - 2, the N-dimensional KAM torus cannot serve as boundaries dividing the energy surface into regions impenetrable for the wandering trajectories. Therefore, when N > 2 these trajectories may wander along the whole energy surface and give rise to a new mechanism of randomness called Arnold diffusion[4]. Physically thls new possibility remains unobservable as far as the KAM conditions hold.

It is appropriate to summarize briefly at this point: Hamiltonian systems with $N = 1$ are all integrable; the overwhelming majority of systems with $N > 2$ becomes nonintegrable; for $N > 2$ Arnold diffusion may show up,

What happens when one violates the conditions of KAM theorem? This appears to be a very difficult problem, just to cite such a competent mathematician as Arnold : "Nonintegrable problems of dynamics appeared inaccessible to tools of modern mathematics"[5]. Still mathematicians wae able to tell the qualitative picture of how the KAM tori are destroyed and modern computers are of much help to visualize this process[6].

Let us try to grasp the essentials without involving mathematics. We start with the mathematical pendulum described by the simple nonlinear differential equation

$$\ddot{\varphi} + \omega^2 \sin\varphi = 0 \quad . \tag{2-9}$$

The angle φ and the angular velocity $\dot{\varphi}$ span a two-dimensional phase plane. Due to periodicity of the motion we can consider only an infinite strip $-\pi < \varphi < \pi$ in the phase plane (Fig, 2-l). Among the equi-energy curves shown in Fig. 2-1, there is one connecting the points $(-\pi, 0)$ and $(\pi, 0)$. This separatrix divides the strip into three regions with different types of motion: oscillation in the central region and rotation in opposite directions in the upper and lower regions.

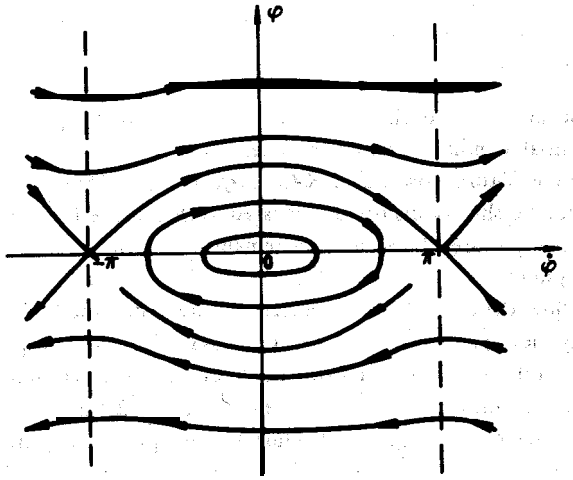


Fig. 2-1 Phase plane of the mathematical pendulum.

In Fig. 2-l the stable equilibrium point (0,0), surrounded by ellipses, is an elliptic point or a center, while the unstable equilibrium point $(\pi, 0)$ is a hyperbolic or saddle point. We see that the separatrix leaves one saddle point along the "unstable" direction and enters the other saddle point along the "stable" direction and vice versa. Actually, $(\pi, 0)$ and $(-\pi, 0)$ correspond to the same point and the "stable" and "unstable" directions of the separatrix intersect at the saddle point.

Now take a system of two coupled oscillators in its nearly integrable regime. A section of the phase space in terms of J_i, Q_i would appear as if assembled from Fig. 2-1, i.e., there are regions corresponding to motion of different frequencies called *resonance zones,* see Fig. 2-2. When the coupling gets stronger and the conditions of KAM theorem begin to break down, these resonance zones tend to overlap and the original separatrices become stochastic layers of finite width (see Fig. 2-3). At the same time, some of the originally simple closed curves split into successions of elliptic and hyperbolic points at small scales. The overlap of resonance zones can be cast into a quantitative criterion for the appearance of stochasticity (the *Chirikov criterion*[7]).

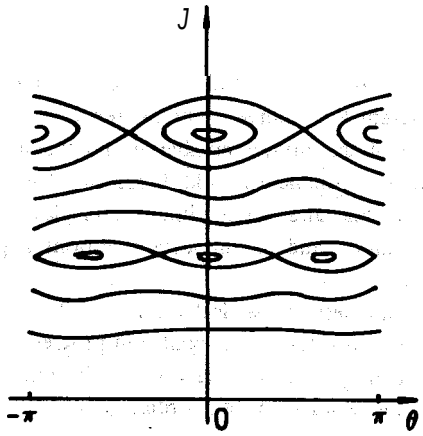


Fig. 2-2 Resonance zones of couples oscillators.

In general, the KAM torus looks much like these fiiures if intersected by planes in the phase space. The KAM curves can be classified according to the "distance" of their underlying frequency ratio from rationals. The more irrational a curve is, the longer it persists during the' violation of the KAM conditions. The destruction of KAM curves shares the general feature of going from Fig. 2-2 to Fig. 2-3. The intersection of stable and unstable directions seen in Fig. 2-1 now generalizes to

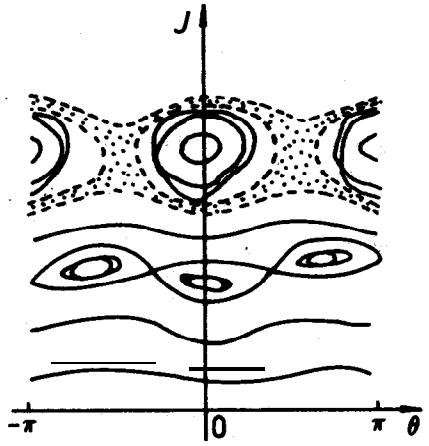


Fig. 2-3 Stochastic layers.

intersections of "stable manifold" and "unstable manifold". A transversal intersection of stable and unstable manifolds is called a *homoclinic point,* if they result from the destruction of one and the same KAM curve, and a *heteroclinic point* if they come from different KAM curves. The existence of a single homoclinic point implies the presence of an infinite number of homoclinic points, because the intersection of stable and unstable manifolds is accomplished in a very intricate way (see Fig. 2-4). Homoclinic and heteroclinic points play the role of organizing centers for chaotic motion and are important notions applicable to dissipative systems as well.

The degree of randomness of classical motion is closely related to its ergodic property, which belongs to a domain of significant progress in last 20 years[8]. Since one encounters the concepts of ergodicity when characterizing the attractors in dissipative systems, a few words may be in order.

A system 'is said to be ergodic on its energy surface if time averages along a trajectory are equal to ensemble averages over the whole energy surface. Ergodicity alone means very little in randomness: two neighbouring points may remain correlated all the time. The next step on the ladder of randomness is called mixing: any initial region on the energy surface evolves into fiients which cover the whole surface when time goes on. Correlation between initial neighbours must decay with time, but nothing is required as regards the decay rate. If the motion suffers local orbital instability, i.e. any nearby trajectories go apart exponentially, then the correlation also decays exponentially. Such system is said to be a *K-flow*

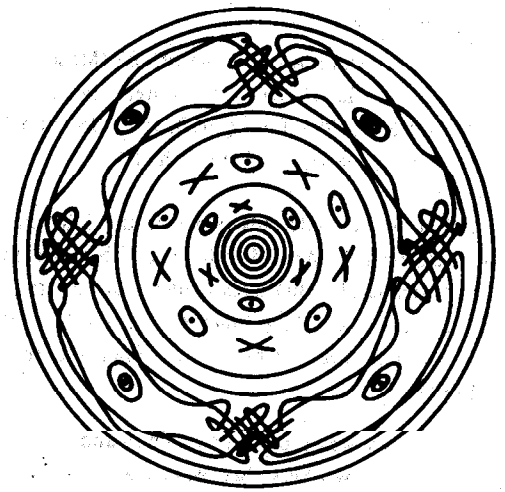


Fig. 2-4 Intersections of stable and unstable manifolds (schematic).

(after Kohnogorov). The separation rate of neighbouring points averaged along the trajectory determines the *K-entropy* of the system. A system is a *K*-flow if it possesses positive K-entropy., This serves as another criterion for stochasticity along with the Chirikov resonance overlap criterion, which often give identical results in simple cases. We' shall return to the notion of entropy in Chapter 8. Skipping the highest level of randomness, namely the Bernoulli flow[8]; we only recall that every higher step on the ladder of ergodicity implies the lower ones, but not vice versa. K-flow is the normal case one encounters in "Hamiltonian chaos".

To conclude this chapter let us emphasize that instability does not mean collapse of the system, but opens the way to intrinsic stochasticity. A deeper thought on the schematic parallel

stability —— determinism

instability —— randomness

would help the reader to get rid of the traditional prejudice that classical mechanics is fully deterministic and to recognize stochasticity as an ubiquitous and intrinsic property of nonintegrable Hamiltonian systems.

References and guide to the bibliography

1. Books on stochasticity in Hamiltonian systems: B2 (Moser, 1973), B4 (Casati and Ford, 1977); B6 (Arnold, 1978), B31 (Lichtenberg and Lieberman, 1982); Reviews: Chirikov (1969; 1979), Ford (1973, 1974, 1975), Whiteman (1977), Berry (1978), Helleman (1980), and Wightman (1981).
2. Siegel (1941, 1954).
3. Kohnogorov (1954). We include this paper as the first one in this volume for its historical significance; Arnold (1963a) and Moser (1962) were purely mathematical, but Arnold (1963b) is readable for physicists.
4. For Arnold diffusion see Arnold (1964), Chirikoy et al. (1979), Tennyson et al. (1979), Lieberman (1980), Chirikov and Shepelyansky (1982), Holmes and Marsden (1982a, 1983).
5. Arnold (1963a).
6. The first computer study was reported by Henon and Heiles (1964) and then continued by Ford and coworkers. See Walker and Ford (1969), Paper 2 in this volume, Ford and Lunsford (1970, 1972), Lunsford and Ford (1972). Renormalization group ideas have been applied to the destruction of KAM surfaces recently, see Kadanoff (1981a, b, 1983a), Escande and Doveil (1981a, b), Shenker and Kadanoff (1982), Escande (1982a, b, c).
7. For stochastic layer and the resonance overlap criterion see the reviews by Chirikov (1969, 1979).
8. An elementary introduction to modern ergodic theory can be found in J. L. Lebowitz and O. Penrose, *Physics Today*, 1973, February, p. 23, but nothing was said about K-systems. We recommend another *Physics Today* paper by Ford, April 1983, p. 40. The standard reference on ergodic theory is B1 (Arnold and Avez, 1968).

Chapter 3
THE PROBLEM OF ONSET OF TURBULENCE

As we mentioned in Chapter 1, turbulence has been a long-standing problem in physics. The difficulty is rooted in the simultaneous presence of many, manylength scales, or in other words, in the lack of a single characteristic length. This can be seen from the intuitive picture of a turbulent fluid: nested and interpenetrated eddies of all scales, from macroscopic down to "molecular". In this respect the problem of turbulence bears similarity to the problem of continuous phase transitions, where length scales ranging from the correlation length, which approaches infinity at the transition temperature, down to the atomic scale all play an essential role. Perhaps this explains why people like L. D. Landau and K. G. Wilson who contributed so much to the understanding of phase transitions also thought about turbulence.

Just as in the case of phase transitions, the key to understanding turbulence may be hidden in the onset mechanism, as pointed out by L. D. Landau forty years ago: " . . . the problem may be in a new light if the process of initiation of turbulence is examined **thoroughly**"[1]. We would like to make it clear from the very beginning that chaotic phenomena in dissipative systems, at least for the time being, are relevant only to the onset mechanism of turbulence, i.e. to the stage of weak turbulence. It has nothing to do with the fully developed turbulence which is of primary importance in engineering. In addition, chaos as it is treated in this volume concerns mainly erratic behaviour in time evolution, whereas turbulence necessarily involves stochasticity in the spatial distribution as well.

Real turbulence occurs in three-dimensional space. However, most mathematical models and experimental situations studied so far are confined to fmite or low-dimensional geometry, e.g., fluid instability between rotating cylinders (the Taylor instability) or thermoconvective instability in small boxes (the Rayleigh-Benard instability). Developed turbulence must involve a great number of fluid motion

modes, but the onset of turbulence may stem from the loss of stability of only a few modes. Finite geometry together with dissipation just provides the mechanism to suppress many irrelevant modes what makes the experimental results closer to predictions based on simple theoretical models.

It has been realized for a long time that turbulence might be a sophisticated regime of nonlinear oscillation in continuous media. L. D. Landau[1] and E. **Hopf**[2] attributed the onset of turbulence to the appearance of an increasing number of quasiperiodic motions resulting from successive bifurcations in the system. Being an extension of the Hopf bifurcation idea, the first steps of this process can easily be understood geometrically.

When the Reynolds' number *R*, which represents the relative importance of nonlinear to dissipative terms in the Navier-Stokes equations (l-l), remains small enough, the fluid motion is laminar and stationary, corresponding to a stable *fixed point* in its phase space. A stable fixed point acts as an *attractor*, i.e., it attracts all nearby initial points towards itself (see Fig. 3-l(a)). Now, let the Reynolds' number be increased infinitesimally larger than the first critical value R_{c1} where the fixed point loses stability and begins to repel all nearby trajectories (Fig. 3-1(b)). Since a small change in *R* cannot cause such drastic consequence as inverting the direction of all flows on the whole phase space, the neighbourhood of the fixed point may become repelling, but it must remain attracting with-respect to regions located far enough. Local repulsion and global attraction of the flow implies the formation of a closed curve around the now unstable fixed point and the curve attracts all nearby

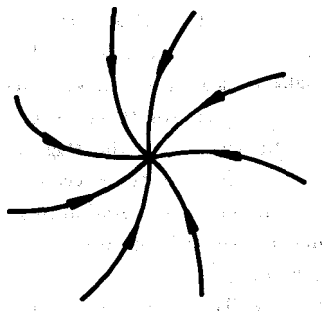


Fig. 3-1 (a)
A stable fixed point.

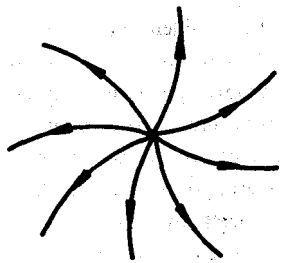


Fig. 3-1 (b)
An unstable fixed point.

flows (see Fig. 3-1(c)). This closed curve is called a *limit cycle* and corresponds to a periodic motion of the system.

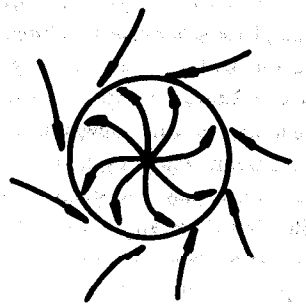


Fig. 3-1 (c)
A compromise appearance of a limit cycle.

The process of generating a limit cycle from a fixed point is called a *Hopf bifurcation.* Repeated use of the above arguments would reveal the nature of the next bifurcation when the limit cycle loses stability and becomes repelling, at R_{c2}. There would appear an attracting closed tube, i.e. a 2-torus, around the unstable limit cycle. The motion becomes quasiperiodic if the two frequencies on the torus are incommensurable. Landau and Hopf allowed this process to continue infinitely and identified the final state with an infinite number of incommensurable frequencies as fully turbulent. This was called the *Landau-Hopf route* (or scenario) to turbulence.

At present, we do not know any reasonable mathematical model which follows the Landau-Hopf route to turbulence. The Landau-Hopf route requires successive appearance of new incommensurable frequencies in the power spectrum, which remains discrete all the time as Reynolds' number increases. Turbulent spectra in Laboratory experiments do develop a few independent frequencies, but then turn into broad noisy bands. There is no mechanism for, sensitive dependence on initial conditions in Landau-Hopf scheme, but the details of turbulent states do depend on initial conditions sensitively, In addition, the Landau-Hopf picture ignores an important physical phenomenon- frequency locking. In fact, in nonlinear systems new incommensurable frequencies cannot appear infinitely without interacting with each other. Nearby frequencies tend to get locked, which will diminish the number of independent frequencies. The above remarks exclude the Landau-Hopf route as an onset mechanism for turbulence from both the theoretical and the experimental points of view.

In 1971 Ruelle and **Takens**[3] showed that the Landau-Hopf route is unlikely to occur in nature. It is enough to have four consecutive bifurcations to get into a state of erratic motion described by interweaving trajectories attracted to a low-dimensional manifold in the phase space called a *strange attractor.* They identifiid the motion on sstrange attractors with turbulence. Their scheme may be summarized as: fixed point → limit cycle → 2-torus → **3-torus** → strange attractor (turbulence). A few years later, in collaboration with Newhouse, these authors succeeded in reducing the scheme to: fixed point → limit cycle → 2-torus → strange attractor, i.e., quasiperlodic motion on a 2-torus may lose stability and give birth to turbulence directly. This so-called *Ruelle-Takens route* to turbulence seems to be more consistent with recent hydrodynamical experiments, but it is less well understood in theoretical models except for some deep results on circle mappings. We shall devote Chapter 7 to transitions from quasiperiodic motion to chaos.

The work of Ruelle and Takens played an eye-opening role. More and more people now believe that at least the problem of turbulence onset can be settled within the framework of Navier-Stokes equations and many new routes to turbulence have been suggested Now we are facing a situation of "all' routes lead to turbulence"; among which the most thoroughly studied are the *period-doubling* route of **Feigenbaum**[4] and the *intermittent route* of **Pomeau-Manneville**[5]. These two routes are actually twin phenomena. The next Chapter 4 on onedlmensional mappings is at the same time an introduction to the period-doubling route and Chapter 6 will deal with the intermittent route, so we shall not go into details here.

In conclusion we would like to point out that turbulence has become a general concept, related to many branches of natural sciences and not less important than the concept of order. New terms such as solid state turbulence, chemical turbulence, acoustic turbulence or optical turbulence are emerging into the literature. Subtle measurements at liquid helium temperatures, laser Doppler velocimetry and modem data acquisition technique[6] have brought the study of turbulence back into physics laboratories. The idea of scaling and universality as well as renormalization group arguments which have proved so successful in understanding phase transitions are now being adapted to chaos and turbulence. In one word, turbulence should not be considered as a specific problem in hydrodynamics. It should attract the attention of physicists, because "That is the central problem which we ought to solve some day, and we have not"[7].

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Chapter 4

UNIVERSALITY AND SCALING PROPERTIES OF ONE-DIMENSIONAL MAPPINGS

It often happens in physics that one-dimensional models are either too trivial or too specific to be extended to higher dimensions, but chaos in dissipative systems offers a lucky and instructive exception. The reason is very simple: dissipation plays a global stabilizing role against local orbital instability and causes the volume representing the initial states in phase space to contract in the process of evolution. This contraction makes the phase volume approach onedimensional objects in some of Its sections and thus enables higher dimensional systems to enjoy the universal properties of one-dimensional mappings.

One-dimensional mappings of an interval into itself are simple enough to be accessible to certain analytical tools and are not very time-consuming in numerical study. At the same time they are rich enough to show many of the scaling and universal properties of chaotic transitions observed in higher dimensional systems. Therefore, we shall treat them in more detail than we did in previous chapters.

Consider a real interval Z and a nonlinear functionfwhich transforms any point x of Z into some point x' in the same interval I. This is called *a map of the interval*

$$f: I \to I \tag{4-1}$$

In general, the function f may depend on a parameter μ, We can choose an arbitrary initial point $x_0 \in$ Z and iterate it using f :

$$x_n = f(\mu, x_{n-1}) \quad , \qquad n = 1, 2, 3, \ldots \tag{4-2}$$

Formula (4-2) can be rewritten as

$$x_n = f^{(n)}(\mu, x_0) \quad , \qquad n = 1, 2, 3, \ldots \tag{4-3}$$

where $f^{(n)}$ denotes the n-th iterate (not derivative!) off and sometimes it is useful

to have another specific notation for it:

$$F(n, \mu, x) \equiv f^{(n)}(\mu, x) \equiv \underbrace{f(\mu, f(\mu, \ldots f(\mu, x) \ldots))}_{n \text{ times}} \quad . \tag{4-4}$$

In general, the property of the sequence $\{x_i, i = 0, 1, 2, \ldots\}$ depends on the function f and on the choice of x_0 and μ. We confine ourselves to those functions f which have only one maximum on the interval I. Without loss of generality one can rescale f and x in such a way that

(1) the maximum is located at $x = 0$, $f'(\mu, 0) = 0$, and $f(\mu, 0) = 1$;

(2) $f(\mu, x)$ is monotonically increasing when $x < 0$, and monotonically decreasing when $x > 0$.

This kind of mappings has been called *unimodal.* We further require that

(3) in the neighbourhood of $x = 0$, f can be expanded as

$$f(\mu, x) = 1 - ax^z + \ldots \tag{4-5}$$

where $z = 2, 4, 6, \ldots$ etc.

For concreteness we shall refer to the *logistic mapping*

$$x_n = 1 - \mu x_{n-1}^2 \tag{4-6}$$

as the representative of unimodal mappings. In this case Z = [- 1, + 1] , $\mu \in (0, 2)$, and z = 2. We see that even when x runs over the whole interval Z, $f(\mu, x)$ does not necessarily fii up the interval Z, hence the name *endomorphism* (endo = internal) of the interval in some mathematical literature.

Now let us return to the sequence $\{x_i, i = 0, 1, 2, \ldots\}$. Usually after a few hundred transient points, it settles into one of two kinds of stationary patterns: periodic or aperiodic. A periodic pattern of period p: $x_{i+p} = x_i$, $x_{i+k} \neq x_i$ for all $k < p$ and all i larger than certain N, is also called an *orbit* of period p or a *p-cycle* for the mapping *f.* In what follows we shall assume that the transients have died away and omit the phrase "for all i larger than certain N". The particular case $p = 1$ corresponds to a fixed point for f:

$$x^* = f(\mu, x^*) \quad . \tag{4-7}$$

By definition, all points from a p-cycle off must be fixed points of the p-th iterate off, i.e.

$$x_i = F(p, \mu, x_i) \qquad i = 1, 2, \ldots, p \quad . \tag{4-8}$$

A standard question to be asked about a fixed point is its stability, i.e. if x_n

is chosen very close to the fixed point x^*:

$$x_n = x^* + \epsilon_n$$

what happens with the next iterate

$$x_{n+1} = x^* + \epsilon_{n+1} \quad ?$$

If

$$|\epsilon_{n+1}/\epsilon_n| < 1 , \tag{4-9}$$

we say that the fixed point x^* is stable, The stability condition (4-9) is equivalent to the requirement

$$|f'(\mu, x^*)| < 1 , \tag{4-10}$$

Similarly, a p-cycle is stable if

$$\left|F'(p, \mu, x_i)\right| = \left|\prod_{j=1}^{p} f'(\mu, x_j)\right| < 1 , \qquad i = 1, 2, \ldots, p , \tag{4-11}$$

where the chain rule of differentiation has been used.

The most favourable case for stability appears when

$$F'(p, \tilde{\mu}, x_i) = 0, \qquad i = 1, 2, \ldots, p , \tag{4-12}$$

then one has quadratic convergence towards the fixed point. The stability condition (4-11) holds for a finite interval called a *periodic window* on the μ-axis, whereas (4-12) takes place only at one particular value $\tilde{\mu}$ somewhere in the middle of the periodic window. This $\tilde{\mu}$ value corresponds to a *superstable period,* which serves as the representative of all p-cycles from the same periodic window. With our conventions on f we can say that any cycle containing the point x = 0 must be superstable, since $f'(\mu, 0) = 0$ leads to $F'(p, \mu, 0) = 0$ for all p. This suggests an idea to determine the superstable value $\tilde{\mu}$ by solving the fixed point equation $F'(p, \mu, 0) = 0$, but we shall mention a better method later.

Equipped with a desk calculator one can easily fmd the first periodic windows for the logistic mapping (4-6):

$p = 1$	$0 < \mu < \mu_1 = 0.75$
$p = 2$	$\mu_1 < \mu < \mu_2 = 1.25$
$p = 4 = 2^2$	$\mu_2 < \mu < \mu_3 = 1.3680989 \ldots$
$p = 8 = 2^3$	$\mu_3 < \mu < \mu_4 = 1.3940461 \ldots$
...	...

This is a *period-doubling bifurcation cascade* with period $p = 2^n$ which quickly converges to an aperiodic orbit at $n = \infty$, the value $\mu_\infty = 1.401155\ldots$ being approached as a geometric progression, namely

$$\mu_n \approx \mu_\infty - \frac{A}{\delta^n} , \qquad \text{as } n \to \infty \qquad A = ? \tag{4-13}$$

where $\delta = 4.66920\ldots$ is a universal (for, unimodal mappings with z = 2) constant, first discovered by M. J. **Feigenbaum**[1].

What happens when μ gets larger than μ_∞ is more interesting from the viewpoint of chaos. In the parameter range $(\mu_\infty, 2)$ there exists an infinite number of periodic windows immersed in the background of aperiodic regime. If one examines more carefully the distribution of points in the aperiodic regime, then one sees the iterates jumping between 2^n subintervals (or islands) of the interval I with n decreasing from ∞ to 0 when μ goes from μ_∞ to 2. This is the so-called *reversed,* or *period-halving bifurcation sequence* of chaotic bands, which looks much like a washed-out mirror image of the direct period-doubling bifurcation sequence with respect to μ_∞. This overall structure can be seen clearly from the bifurcation diagram, obtained by plotting a few hundred stationary outputs of the iteration (4-6) versus the parameter μ (see[2] for figures).

If what described above is restricted only to a specific mapping, e.g. (4-6), it would be worth no more than a rare bird in the mathematical zoo. However, this kind of bifurcation structure with its universal numerical characteristics (such as the Feigenbaum constant δ) appears more and more in nonlinear mathematical models and real experiments. In this introductory chapter we confine ourselves to listing some useful notions and rigorous results related to the universal and scaling properties of one-dimensional mappings.

1. A unimodal mapping can have at most one stable period for each parameter value μ. It may have no stable period at all for many μ values. The necessary condition for f to have at most one stable period was found in 1978 and consists in the *Schwarzian derivative Sf* of f being negative on the interval I[3]:

$$Sf(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2}\left(\frac{f''(x)}{f'(x)}\right)^2 < 0 \tag{4-14}$$

(see[4] for more on the Schwarzian derivative). The condition (4-14) is not sufficient for the stable period to exist. It is just this insufficiency that opens the possibility for chaotic orbit to appear: even when $Sf < 0$ one can get different aperiodic sequences starting from different x_0 and never reach a stable period.

2. The set of all parameter values which give birth to chaotic orbits possesses a positive measure on the μ-axis, although no such μ values form an interval. Moreover, to classify an orbit as chaotic one must show that all consecutive points approach a continuous distribution with respect to dx. Computers are of no use in proving this kind of statements, and one must appeal to rigorous mathematics. Proofs for some particular parameter values have been known since long[5], but a general proof for certain classes of mappings appeared only in 1981[6].

3. The classification and ordering of periodic orbits has been a well-studied mathematical problem with an extensive literature. In this and the next subsections we list a few results, frequently encountered in the literature. First of all, there was a theorem proved in 1964[7] which became known to physicists much later:

Sarkovskii theorem: consider the following ordering of integers

$$3 \to 5 \to 7 \to 9 \to \ldots \to 3*2 \to 5*2 \to 7*2 \to 9*2 \to \ldots \to 3*2^2 \to 5*2^2$$

$$\to 7*2^2 \to 9*2^2 \to \ldots \to 3*2^n \to 5*2^n \to 7*2^n \to 9*2^n \to \ldots\ldots 2^m$$

$$\to \ldots \to 32 \to 16 \to 8 \to 4 \to 2 \to 1 .$$

(the symbol $\to$ means "precede"). If f is an unimodal mapping and has a point x leading to a p-cycle, then it must have a point leading to a q-cycle for every $q \leftarrow p$ in the sense of the above ordering.

It must be emphasized that the Sarkovskii theorem is only a statement concerning different x's at fixed parameter value μ. It says nothing about the stability of the periods, nor about the measure, i.e. the observability of these periods. Unaware of the work of Sarkovskii, many equivalent formulations or corollaries of this theorem were suggested in the intervening years, among which the Li-Yorke theorem "Period 3 implies chaos"[8] and the Oono's statement "Period $\neq 2^n$ implies chaos"', etc.

4. The *U-sequence* and *symbolic dynamics.* The order of appearance of various stable periods hi an unhnodal mapping $f(\mu, x)$ when the parameter μ varies is certainly more important for physics. We'introduce here the notion of *U-sequence,* called also *MSS* sequence in the literature according to the authors of[10] or kneading sequence according to the procedure how it is constructed.

With our conventions on the scale of $f(\mu, x)$ and x in mind, let us take $x = 0$ as the "Centre" and initial point of the mapping and then follow the subsequent iterations. We mark a point by the letter R or L according to whether it falls to the Right or Left of the Centre. Therefore, each periodic sequence corresponds to a word of finite length made of R's and L's. Then the important result of *MSS*[10] says that it is possible to introduce an ordering for all admissible words, i.e. to

compile a dictionary of these words and there is a correspondence between the position of a word in the dictionary and the place of occurrence of a period on the parameter axis for a wide class of unimodal mappings. In particular, if the word P corresponds to a period, we can construct another word $H(P) = PxP$, where $x = R$ if there is an even number of R's in P and $x = L$ otherwise. H is called the harmonic of P and represents the period-doubled orbit adjacent to P. All admissible words corresponding to periods equal to or less than 11 were given in the Appendix of[10].

Acutally the construction of harmonics happens to be a particular case of a more general composition rule, introduced and denoted by * in the paper of Derrida *et al.*[11]. Using the * composition one can express the harmonic of P as $P*R$, and a period-doubling sequence starting with P corresponds to the symbolic sequence $P*R^{*n} = P*R*\ldots\ *R$ (n times of R), etc. Moreover, one can select period-tripling sequence $(RL)^{*n}$, period-quadrupling sequence $(RL^2)^{*n}$, period-quintupling sequences $(RLR^2)^{*n}$, $(RL^2R)^{*n}$ and $(RL^3)^{*n}$, etc. from the infinite number of periods embedded in the chaotic bands. Periods in these sequences are not adjacent on the parameter axis, but they have their own scaling factors and convergence rates.

Exploring the properties of the * composition, Derrida et *al.*[11] succeeded in proving the self-similarity of the U-sequence, i.e. a part of it appears to be similar to the whole sequence. This important result is-the mathematical manifestation of the hierarchy structure of chaotic bands, seen in bifurcation diagrams. *U*-sequence has been observed in real[12] and computer experiments[13].

Classification of periods by using combinations of two letters is an application of symbolic dynamics, a notion which originated in topological dynamical systems theory in the 30's and later adapted to differential dynamical systems by Bowen and others;. For physicists, symbolic dynamics is nothing but a mathematical realization of coarse-graining. Instead of detailed information on the phase trajectory at all times, one simply labels a reasonable partition of the phase space by a set of letters and then follows the succession of letters. If properly chosen, this symbolic dynamics will reflect the essential feature of the evolution process, e.g. its periodicity. The method of symbolic dynamics is expected to play more role in physics (see the Preface of J. Ford to Ref. 14).

5. How to determine the superstable parameter values? Again with our conventions on f in mind we define a sequence of polynomials $P_n(\mu)$ using recursive relations based on the original mapping $f(\mu, x)$:

$$P_0(\mu) = 0 \ ,$$

$$P_n(\mu) = f(\mu, P_{n-1}(\mu)) \quad . \tag{4-15}$$

The $P_n(\mu)$'s as functions of μ describe the dark lines seen in the chaotic region of bifurcation diagrams (see figures cited in Ref. 2), because the iterates of $x = 0$ always correspond to extremes in islands of the chaotic bands. Actually, $P_n(\mu)$ is nothing but the n-th iterate of ftaken at point $x = 0$, i.e.

$$P_n(\mu) \equiv F(n, \mu, 0) \quad . \tag{4-16}$$

Therefore, real zeros of $P_n(\mu)$ in the appropriate interval give the parameter values for superstable orbits of period n. For example, $P_5(\mu)$ in the case of the logistic mapping (4-6) has three zeros 1.62541 . . . , 1.86078.. . and 1.98542 . . . , which correspond to superstable orbits of types RLR^2, RL^2R and RL^3. However, when zeros get very close to each other most numerical algorithms to solve $P_n(\mu) = 0$ lose stability, which makes this method impractical. Nevertheless, there exists a nice method to calculate the superstable parameter value separately; given the corresponding word in the U-sequence. Take for example the word RLR^2 which represents the first superstable period 5. Write it as a 5-cycle starting and ending at $x = 0$, i.e.

$$f_R(\mu, f_L(\mu, f_R(\mu, f_R(\mu, f(\mu, 0))))) = 0 \tag{4-17}$$

where the subscript R or L indicates which branch, Right or Left, of the map has been used at a given step. (The first f maps into the Centre and thus has no subscript.) Once the branches have been marked, we can invert (4-17) to yield

$$1 = f(\mu, 0) = f_R^{-1}(f_R^{-1}(f_L^{-1}(f_R^{-1}(0)))) \quad . \tag{4-18}$$

Now define

$$R(x) = f_R^{-1}(x), \qquad L(x) = f_L^{-1}(x) \tag{4-19}$$

and rewrite (4-18) using the, function composition symbol $\circ$. We see that the word RLR^2 has been "lifted" to an equation for μ:

$$R \circ L \circ R \circ R(0) = 1, \tag{4-20}$$

which determines the superstable parameter μ for RLR^2, In the case of the logistic map (4-6) taking the square of (4-18) a few times to get rid of the radicals would bring us back to $P_5(\mu) = 0$ with all other zeros present. So we retain (4-20) and multiply it by μ to get

$$\sqrt{\mu + \sqrt{\mu - \sqrt{\mu - \sqrt{\mu}}}} = \mu$$

H. Kaplan[15] suggested to solve this equation by iteration, i.e. let

$$\mu_{n+1} = \sqrt{\mu_n + \sqrt{\mu_n - \sqrt{\mu_n - \sqrt{\mu_n}}}}$$

with a suitable μ_0, say, $\mu_0 = 2$. One can easily read out the general rule to write down the iteration equation for any word in the U-sequence.

6. The renormalization group *(RG)* equation and the scaling factor. Generally speaking, behind any *RG* arguments there always figures some geometry with infinitely-nested self-similar internal structure (a kind of fractal geometry, to use a term now in fashion). One-dimensional mappings provide us with a simple example.

The very instructive discussion on scaling property of one-dimensional mapping in Feigenbaum's first paper was illustrated by several not well-proportioned figures, so we redraw them in Fig. 4-1. Only the vicinity of certain fixed points is shown in these figures. A part of $f^{(2^n)}$ at the exact superstable parameter μ_n is given in Fig. 4-1(b). If we draw $f^{(2^{n-1})}$ at the same parameter value μ_n, it represents the period-doubled regime shown in Fig. 4-l(a), where locally one has a few superstable 2-cycles as outlined by the square boxes. Now increase μ_n from the situation of Fig. 4-1(b), until a new period-doubled superstable regime appears at the next parameter, value $\mu_n + 1$, as shown by the smaller boxes in Fig. 4-l(c). Comparing the two hatched boxes in Figs. 4-l(c) and 4-l(a) shows what happened was a rescaling and change of signs in both the x and y directions. This process repeats itself with μ increasing.

To put the above geometrical observation into a mathematical frame, let us introduce an operator T to represent this period-doubling, μ-shifting and rescaling procedure starting from some fixed μ_n, i.e.

$$\begin{aligned} Tf(\mu_n, x) &= -\alpha f(\mu_{n+1}, f(\mu_{n+1}, -x/\alpha)) = -\alpha F(2, \mu_{n+1}, -x/\alpha) \\ T^2 f(\mu_n, x) &= (-\alpha)^2 F(2^2, \mu_{n+2}, x/(-\alpha)^2) \end{aligned} \tag{4-21}$$

Feigenbaum gave some plausible. arguments to conjecture the existence of an universal, i.e. independent on the starting f, limit

$$\lim_{k \to \infty} (-\alpha)^k F(2^k, \mu_{n+k}, x/(-\alpha)^k) = g(x) \tag{4-22}$$

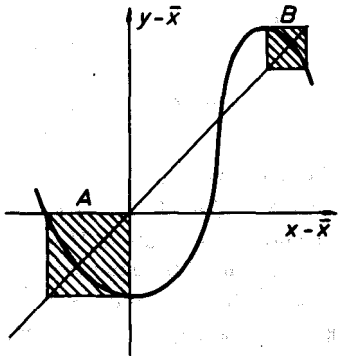


Fig. 4-1(a)
A segment of $f^{(2^{n-1})}(\tilde{\mu}_n, x)$, $\tilde{\mu}_n$ is the superstable point for $f^{(2^n)}$.

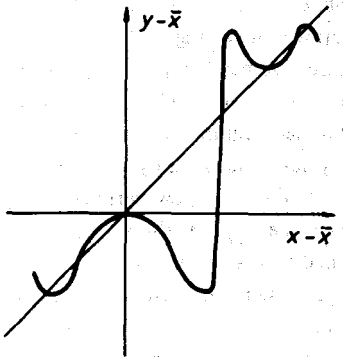


Fig. 4-1(b)
A segment of $f^{(2^n)}(\tilde{\mu}_n, x)$.

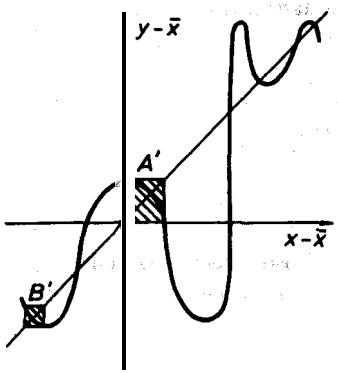


Fig. 4-1(c)
A segment of $f^{(2^n)}(\tilde{\mu}_{n+1}, x)$ to be rescaled and inverted to get (a) again.

This conjecture was proved later[16]. From the definition (4-4) of the composite function F it follows

$$F(2^k, \mu_{n+k}, x/(-\alpha)^k) = F(2^{k-1}, \mu_{n+k}, F(2^{k-1}, \mu_{n+k}, x/(-\alpha)^k)) \quad (4\text{-}23)$$

Taking the $k \to \infty$ limit on both sides of (4-23) results in the *RC* equation of Feigenbaum:

$$g(x) = -\alpha g(g(-x/\alpha)) \quad (4\text{-}24)$$

The normalization of $f(\mu, 0) = 1$ and the superstable condition (4-12) imply the following boundary conditions for (4-24):

$$g(0) = 1 \ , \ g'(0) = 0 \quad (4\text{-}25)$$

It is worth mentioning that conditions (4-25) alone do not determine the solution of (4-24) uniquely. At least the behaviour of g(x) near x = 0. should be given, e.g. series expansion

$$g(x) = 1 + Ax^z + Bx^{2z} + Cx^{3z} + \ldots \quad (4\text{-}26)$$

will give different solutions to (4-24) for different choice of z .

The *RG* equation (4-24) can be generalized to period-*n*-tupling sequences:

$$\alpha^{-1} \ g(\alpha x) = -g^{(n)}(-x) \quad (4\text{-}27)$$

Equation (4-24) defines a saddle point in the space of all functions leading to $g(x)$[17]. Therefore, one must be careful in devising numerical procedure to calculate g(x), because the approximations may get away from the true solution after converging to it initially. This situation resembles the *RG* theory of continuous phase transitions where the critical point also corresponds to an unstable saddle point.

Actually the simplest way to solve equations (4-24) or (4-27) consists in substituting the expansion (4-26) and then solving the resulted equations for α, A, B, ... etc. by combined use of algebraic manipulation and numerical languages[18].

7. Convergence rate δ and other universal onstants. The Feigenbaum constant $\delta = 4.66920\ldots$ only determines the convergence rate of period-doubling bifurcation sequences for quadratic mappings, i.e. for z = 2 in (4.5). There is a smooth dependence of δ on z and its values for z= 4, 6, 8 are 7.248, 9.296 and 10.048 respectively[9]. In general, δ is defined as the eigenvalue in excess of 1 of a linearized functional equation obtained from the *RG* equation. Therefore, different solutions of the *RG* equation lead to different α's and δ's for various period-*n*-tupling sequences[18].

There are many other universal constants related to one-dimensional unimodal mappings. Not being able to go into details in this Introduction, we simply list some of them: the height ratio of two adjacent period-doubled peaks in the power spectra is 13.2 db[20], the critical slowing down exponent at μ_n is $\Delta = 1$ [21], the fractal dimension of the attractor at μ_∞ equals 0.538 . . .[22] etc. Another exponent related to external noise will be mentioned in Chapter 8.

8. The crisis of chaotic attractors[23]. The "microscopic" structure of a chaotic attractor does not depend continuously on parameter μ, whereas the general shape of the attractor changes smoothly with μ except for certain values where abrupt changes take place. This phenomenon was called crisis of the attractor and explained by Grebogi et al.[23] as the result of collisions of the attractor with an unstable orbit. Actually all the band-merging points in the bifurcation diagram occur at intersections with unstable orbits left by the main period-doubling bifurcations from the other side of μ_∞.

The sudden change or disappearance of chaotic attractors at "collisions" with other unstable objects certainly plays a more important role in higher dimensional systems.

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Chapter 5
BIFURCATION AND CHAOS IN HIGHER DIMENSIONAL SYSTEMS

Most physical processes are described by ordinary or partial differential equations, so we are more concerned with chaotic behaviour in such higher dimensional systems. Discrete mappings of a plane or an annulus occupy an intermediate place and often provide the clues for understanding chaotic transitions in higher dimensional systems. In what follows we first list some mathematical models encountered frequently in the current literature. Then we shall say a few words about analytical and numerical methods to study these systems, with emphasis on the latter, since one has to rely heavily on numerical experiments in studying higher dimensional systems. A table summarizing the findings in computer experiments will be given at the end of this chapter.

Higher dimensional systems may be divided into three categories: discrete mappings of two or more dimensions (difference equations of second order or higher); ordinary differential equations (ODE's, autonomous, nonautonomous and time-delayed); and partial differential equations (PDE's). We cite examples from each category.

1. Two- and higher-dimensional mappings

In contradistinction to one-dimensional mappings, higher-dimensional mappings may be conservative (volume-preserving) as well as dissipative (volume-contracting), invertible as well as noninvertible, depending on the parameter range in the model. As the first example one should mention the well-studied *Henon mapping*[1]*:*

$$x_{n+1} = 1 - \mu x_n^2 + y_n$$

$$y_{n+1} = b\, x_n \qquad (5\text{-}1)$$

with the Jacobian

$$J = \frac{\partial(x_{n+1}, y_{n+1})}{\partial(x_n, y_n)} = -b \quad .$$

It is an invertible transformation when $b \neq 0$ and goes back to the logistic mapping (4-6) when $b = 0$. It preserves area for $b = 1$ and corresponds to a dissipative system for $b < 1$. The case $b = 0.3$ was studied in great detail by Henon and many others. The essential difference from one-dimensional mappings may be summarized as follows. First, even for one and the same parameter value, the character of stationary output of (5-l) depends on initial point taken. In other words, the (x, y) plane divides into basins, and starting from different basins one may be led to different periodic or aperiodic orbits. In contrast, there exists at most only one stable orbit for one-dimensional unhnodal mappings. Second, for certain parameters and initial points, the interactions of (5-l) converge to an attractor with self-similar internal structure (see Figs. 3-6 in Paper 13). This was the first strange attractor known to have fractal dimension. We shall return to this point in Chapter 9. Third, it was proved that there exist intersections of stable and unstable manifolds in Henon's model, i.e. homocllnic points give rise to chaos.

Another well-known two-dimensional mapping is the so-called *standard mapping*[2]:

$$x_{n+1} = x_n + y_{n+1}$$

$$y_{n+1} = y_n - \frac{\mu}{2\pi} \sin 2\pi x_n \quad . \qquad (5\text{-}2)$$

Being an example of two-dimensional Hamiltonian system, this conservative mapping occurs in many physical applications such as the, motion of charged particle in toroidal magnetic field, or the Frenckel-Kontorova model of modulated structures in solid state physics (see Chapter 10).

Instead of listing other two- and hlgherdhnensional mappings we confine ourselves to a few remarks. First, some mathematical results on one-dimensional mappings can be "lifted" to mappings on R^n as outlined in Ref. 4. Higher dimensional mappings are richer in their behaviour, e.g., basin dependence and transitions from quasiperiodic motion to chaos may take place. Second, period-doubling bifurcations do occur in area-preserving mappings of the plane, but the universal convergence rate happens to be δ = 8.7210. . .[5] . Third, complex mappings[6] present a new class of two-dimensional mappings. While no physical relevance is known for the time being, their graphic representations are beautiful creations of computer science.

2. ***Ordinary differential equations***

ODE's displaying chaotic behaviour may be subdivided into three groups: autonomous systems with three or more variables, nonautonomous systems with two or more variables, and time-delayed systems of at least one variable.

A system of ODE's is called *autonomous,* when there is no explicit time dependence on the right hand side of the standard form:

$$\frac{dx_i}{dt} = f_i(x_1, \ldots, x_N) \ , \qquad i = 1, \ldots, N. \tag{5-3}$$

The classical example of an autonomous system exhibiting chaos is the ***Lorenz model***[7]***:***

$$\begin{aligned} \dot{x} &= -\sigma(x-y) \ , \\ \dot{y} &= -xz + rx - y \ , \\ \dot{z} &= xy - bz \ . \end{aligned} \tag{5-4}$$

It was obtained by truncating the PDE's describing thermal convection between two Infinite plates. There are three parameters in this model: r is the ratio of the Rayleight number to its first critical value, σ is the Prandtl number, and b.

The Lorenz model is one of the simplest truncated hydrodynamical systems, If one considers planar flow of a fluid with periodic boundary conditions imposed in both directions, then the original PDE's (the Navier-Stokes equations) can be transformed into an infinite system of ODE's for the Fourier coefficients of various hydrodynamical quantities. By truncating this latter system one gets various finite systems of ODE's. Truncated equations with four, five, six, seven or more modes retained have been suggested[8], among which the following two-parameter family of seven-mode equations[9]

$$\begin{aligned} \dot{x}_1 &= -2x_1 + 4\sqrt{5}\,x_2x_3 + 4\sqrt{5}\,x_4x_5 \\ \dot{x}_2 &= -9x_2 + 3\sqrt{5}\,x_1x_3 \\ \dot{x}_3 &= -5x_3 - 7\sqrt{5}\,x_1x_2 + 9\epsilon x_1x_7 + R \\ \dot{x}_4 &= -5x_4 - \sqrt{5}\,x_1x_5 \\ \dot{x}_5 &= -x_5 - 3\sqrt{5}\,x_1x_4 + 5\epsilon x_1x_6 \\ \dot{x}_6 &= -x_6 - 5\epsilon x_1x_5 \\ \dot{x}_7 &= -5x_7 - 9\epsilon x_1x_3 \end{aligned} \tag{5-5}$$

permits a continuous transition from a five-mode ($\epsilon = 0$) to a seven-mode ($\epsilon = 1$) model, both well-studied in the literature[8].

The behaviour of all the above-mentioned models is very complicated, so Rossler tried to construct models as simple as possible, but still exhibiting chaotic behaviour. The simplest of his models contains only one quadratic nonlinear term[10]:

$$\begin{aligned} \dot{x} &= -(y+z) \\ \dot{y} &= x + ay \\ \dot{z} &= b + xz - cz \end{aligned} \tag{5-6}$$

Among other autonomous systems displaying chaotic behaviour we mention the model of coupled unstable and damping waves in plasma", the double -diffusive or two-component Lorenz model[12], a 32-mode system supposed to describe the Taylor instability of rotating fluid[13], the 40-mode system for Gunn instability in semiconductors[14], and the coupled Brusselators[15]. (The Brusselator is the name for a hypothetical model of trimolecular reaction with an autocatalytic step, which shows rich temporal and/or spatial structures when a diffusion term is added[16].)

We digress to explain why at least three variables are required to allow chaotic behaviour. This can best be seen from the example of period-doubling as the first step leading to chaos. Since the trajectory has no reason to change drastically at an infllesimal increase of the parameter value, and yet the period T doubles to $2T$, the only imaginable picture is an almost imperceptible splitting of the original orbit. If this splitting takes place in a plane, then there must be at least one point where the trajectory intersects itself and thus violates the uniqueness of the solution. However, usually the ODE systems under consideration are good enough to ensure the validity of the uniqueness theorem. Therefore, splitting of the orbit without self-intersection can take place only in three- and higher-dimensional space: The subsequent bifurcations and the development of chaotic trajectories follow the same principle.

Now let us return to *nonautonomous* ODE's. It is well-known that a non-autonomous system can be made autonomous by adding one or more variable, so at least two variables are required for a nonautonomous system to show chaotic behaviour: In fact, the only kind of nonautonomous systems studied in detail up to now consists of nonlinear oscillators driven by external periodic force.

The forced van der Pol equation[17]

$$\ddot{x} - k(1-x^2)\dot{x} + x = b\lambda k \cos(\lambda t + \varphi) \tag{5-7}$$

seemed to be the first example exhibiting stochastic behaviour reported long before the modern jargon of chaos and strange attractor has emerged.

The forced anharmonic oscillator

$$\ddot{x} + k\dot{x} - \beta x + \alpha x^3 = b \cos \omega t \ , \tag{5-8}$$

being a particular case of the Duffing's equation's

$$\ddot{x} + k\dot{x} + f(x) = g(t) \tag{5-9}$$

where g(t) is a periodic function of t, and f(x) a nonlinear function of x, appears in many physical applications, e.g., motion of dislocation line in supersonic field".

The forced mathematical pendulum

$$\ddot{x} + k\dot{x} + \sin x = a \cos \omega t \tag{5-10}$$

and the parametrically excited **pendulum**[20]

$$\ddot{x} + k\dot{x} + (A + \alpha \cos \omega t) \sin x = 0 \tag{5.11}$$

represent another class of physical models, encountered, say, in describing Josephson junctions In microwave cavity (see References at the end of Chapter 10).

The forced **Brusselator**[21]

$$\begin{aligned} \dot{x} &= A - (B+1)x + x^2 y + \alpha \cos \omega t \\ \dot{y} &= Bx - x^2 y \end{aligned} \tag{5-12}$$

provides at the present time the most thoroughly studied model by combined use of various numerical, methods. All well-known routes to chaos and the U-sequence of Metropolis-Stein-Stein (see Chapter 4) have been shown to exist in this model.

In principle, every autonomous system can be extended to a nonautonomous one by adding a periodic force. From such models we mention only the forced Lorenz **model**[22].

Most periodically forced systems can be viewed as coupled systems of one nonlinear and one linear oscillators. This point of view allows for a more Intuitive interpretation of various regimes of oscillation, Including period-doubling and chaos. Having the driving frequency as a control parameter at hand opens the possibility to reach very high frequency resolution by using various stroboscopic sampling techniques (see below). These merits distinguish the periodically driven equations from purely autonomous systems in numerical studies.

From recent studies on forced nonlinear oscillators there is a short paper worthy of **note**[23]. An exactly solvable nonlinear oscillator was subjected to periodic kicks.

Using the known solution of the free oscillator this model was transformed into a discrete mapping of two variables which enables a detailed study using a reasonable amount: of computer time. Further investigation on this model is desirable, because there is a good chance of discovering U-sequences similar to that observed in the forced Brusselator and of exploring how the Faray sequences describing locking frequencies Immersed in quasiperiodic regime are replaced by the U-sequences describing periodicities embedded in the chaotic region when nonlinearity gets stronger. We believe these. issues are universal for a wide class of nonlinear systems and are relevant to the transition from quasiperiodicity to chaos.

Now comes the last group of ODE's — the *time-delayed equations.* A seemingly simple equation like

$$\tau \dot{x}(t) + x(t) = \mu x(t-1)(1 - x(t-1)) \tag{5-13}$$

may possess very complicated bifurcation and chaos structure except for the limiting case $\tau \ll 1$ when it can be approximated by the logistic mapping (4-6). Formally the time delay may be written as an infinite order differential operator acting on the function x(t) :

$$x(t-1) = \exp\left(-\frac{d}{dt}\right) x(t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{d^n}{dt^n} \, x(t)$$

Now it becomes clear that a time-delayed equation in one variable may correspond to an infinite system of ODE's. One meets this kind of equations In models describing chaos in optical bistability devices (see references at the end of Chapter 10).

3. *Partial differential equations*

Many ODE's listed above are truncations of PDE's. However, direct study of chaotic behaviour in PDE's is still in its beginning stage. We mention briefly a few lines of research on this topic.

Numerical study of PDE's is a time-consuming task. A recent paper reported the first example of period-doubling and chaos In a system of PDE's for thermosolutal **convection**[24]. Actually the equations in (t, x, z) were replaced by a finiie difference scheme on a mesh and the essential nature of the solution was shown not to depend upon altering the mesh interval. Oscillations, period-doublings and chaos were observed in the time evolution of certain global characteristics of the solution. Nothing essentially new was reported compared to ODE's, although one should expect some new features in chaotic behaviour in PDE's, e.g. the interplay between spatial structure and temporal chaos.

Another interesting development connects the two extremes of nonlinear equations: those exhibiting chaos and those having solitons. It is well-known that PDE's solvable by the inverse scattering technique are integrable systems,. The existence of an infinite number of integrals of motion guarantees the creation of such "stable" objects as solitons and precludes the possibility for chaotic behaviour. However, mtegrability is a very subtle property which can easily be broken by periodic perturbations. Chaotic behaviour has been observed in the perturbed sine-Gordon equation[25] the perturbed nonlinear Schrodinger **equation**[26] and the closely related Ginsburg-Landau **equation**[27]. In particular, the interplay of coherent spatial structure and temporal chaos and the evidence for spatial period halving reported in Ref. 25 may have far-reaching consequence, e.g., in understanding cell division in developmental biology.

Now we turn to the methodological aspects in studying higher dimensional systems. As regards analytical tools, there are very few mathematical results ready for physicists to use: no *a priori* criterion for chaos, no classification and enumeration theorem for attractors, etc. Still we would like to call attention to two methods due to Melnikov and Silnikov respectively, both centered on the existence of homoclinic points or homoclinic orbits.

The ***Melnikov's method***[28] applied to near-integrable systems subjected to dissipative time-dependent (usually periodic) perturbations. We mentioned in Chapter 2 that a separatrix becomes a stochastic layer under perturbation which makes the system nonintegrable, but still Hamiltonian. This is not the general rule for dissipative perturbations; as it can be seen from Fig. 5-1. An unperturbed separatrix shown in (a) may be perturbed into one of other three cases, among which only case (d) leads to an infinite number of homoclinic intersections. The criterion for case (d) consists in the alternating sign of the Melnikov function – the distance between stable and unstable branches of the trajectory.

The ***Silnikov's method*** applies to stationary points of saddle-focus type, when there is a homoclinic trajectory passing through the saddle-focus. The theorem says roughly that if the only real eigenvalue and the real part of a pair of complex eigenvalues of the linearized system are of opposite sign and the absolute value of the former is larger, then there exists a set of chaotic trajectories near the original homoclinic trajectory. This criterion has been used for the Rossler model (5-6) and a few other systems.

Before going to numerical methods we discuss a plausible objection against any numerical study of chaos using digital computers. The point is since one always works with a finite field of rational numbers representable on a digital computer of finite word length, it is impossible to realize even a true quasiperiodic process, not to mention chaotic orbits. Actually, things are not so hopeless.

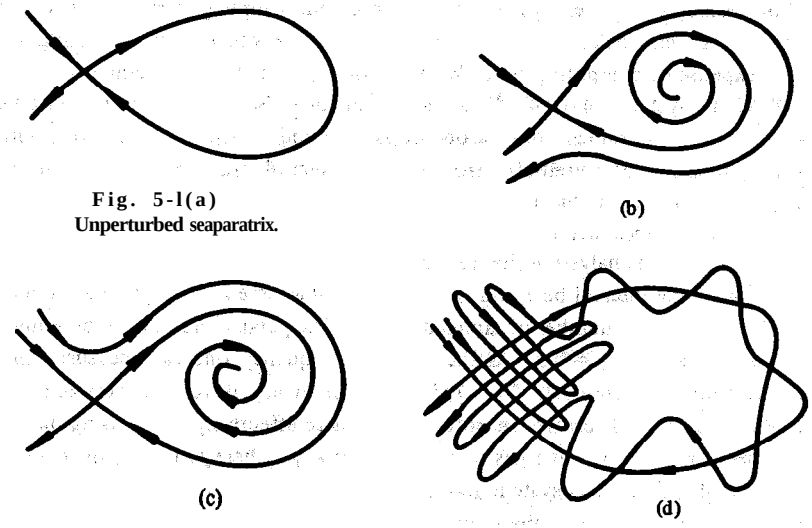


Fig. 5-1(a)
Unperturbed seaparatrix.

Fig. 5-1(b), (c) & (d)
Perturbed sparatrices in dissipative system.

Irrational numbers can be approximated by rationals, chaotic regionsare surrounded by periodic regimes. Our strategy in computer study consists in identifying the periodic orbits with confidence and then characterizing the "aperiodic" motion. We postpone the problem of characterizing the attractors to Chapter 8 and concentrate on how to reach high frequency resolution in numerical experiments. We shall compare the resolving power in terms of the recognizable order p of subharmonics with respect to the natural fundamental frequency of the system under study: $p = 2^n$ for period-doubling sequences, etc.

Direct observation of the trajectories is a method with the least resolution, but it enjoys the merit of having more physical intuition. In this way it is difficult to resolve subharmonics higher than $p = 32$. Sometimes it is useful to draw the stereoprojections of a trajectory in some three-dimensional subspaces of the phase space.

The plotting of Poincare maps, i.e. the intersection points of a given trajectory with a fixed surface in the phase space, offers an effective means to explore the nature of the motion. A practical problem consists in matching the precision of the interpolation scheme used to locate the intersection point with that of the integration algorithm. In this respect a clever suggestion by **Hénon**[30] deserves to be mentioned.

For periodically driven systems the stroboscopic sampling at fundamental and subharmonic frequencies provides the means to reach very high frequency resolution at the expense of computing time. We identified period-doubling sequences up to p = 8192 in systems of ODE's[31] using this method. Some precautions must be taken in using the subharmonic stroboscopic sampling method, e.g., to cope with the accumulation of round-off errors or to distinguish transient or intermittent behaviour from chaotic behaviour. The interested reader may consult the original papers[31] for technical details.

Power spectrum analysis in the frequency domain deserves a few more comments. Let the sampling interval be τ and the total sampling time for a single spectrum be $L = N\tau$, N being the number of sampled points. The parameters τ and L determine two frequencies: $f_{max} = 0.5/\tau$ is the maximal frequency. one can measure using the given sampling interval; $\Delta f = 1/L$ is the frequency difference between two adjacent Fourier coefficients In order to eliminate effectively the aliasing phenomena (see, e.g., Ref. 32) one has to take $f_{max} = kf_0$, where f_0 is the fundamental frequency of the physical system and k is a multiplier of the order of 4 to 8. We aim at resolving the p-th subharmonic of f_0 and wish the subharmonic peak to be formed by s points in the spectrum, i.e., $f_0/p = s\Delta f$. Putting together all the above-mentioned relations, we get

$$p = N/(2ks) . \qquad (5\text{-}14)$$

Note that this is a relation independent on τ and f_0. Taking $N = 8192$, $k = 4$, $s = 8$, we have $p = 128$. This is the lhnit of resolution in power spectrum analysis on most medium-size computers. Nevertheless, power spectrum analysis remains the most useful means to tell different periods embedded in the chaotic bands and the presence of broad-band noise in the spectra is still the most practical symptom for chaos in computer and laboratory experiments.

We close this chapter by summarizing the main findings in computer experiments on various models in the following table.

Table Brief summary of computer experiments on differential equations

Model	Method	Findings	Reference
Lorenz (5-4)	Poincare map Power spectra Lyapunov exponents Dimension	Period-doubling Intermittency Transient chaos Strange attractor	7
Forced Brusselator (5-12)	Stroboscopic sampling Power spectra Lyapunov exponents Dimension	Period-doubling Intermittency Quasiperiodic to chaos Hierarchy of chaotic bands U-sequence and attractor	21
Rossler (5-6)	Power spectra Dimension	Period-doubling	10
Coupled Brusselator	Poincare map Lyapunov exponents Dimension	Period-doubling Intermittency Metastable chaos Quasiperiodic to chaos Strange attractor	15
Double-diiusive convection	Trajectory	Period-doubling	12
3-wave coupling	Poincare map Power spectra Lyapunov exponents Dimension	Period-doubling Strange attractor	11
5-mode truncat. Nader-Stokes	Poincare map Power spectra	Period-doubling	8
6-modetruncat. Navier-Stokes	Poincare map Power spectra	Quasiperiodic to chaos Hysteresis, Intermittency	8
7 -mode truncat. Navier-Stokes	Poincairemap	Quasiperiodic to chaos Hysteresis	8
Duffing (5-9)	Poincare map Powerspectra	Period-doubling Quasiperiodic to chaos Strange attractor	18
Paramaetric pendulum	Stroboscopic sampling Power spectra Lynapunov exponents	Period-doubling Strange attractor	20
PDE's of thermo-solutal convection	Trajectory Timeevolution of Nusselt numbers	Period-doubling Chaos	24

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Chapter 6
THE INTERMITTENT TRANSITION

The term intermittency has been used in the hydrodynamical theory of turbulence to denote random burst of turbulent motion on the background of laminar flow[1]. However, its mechanism slightly differs from what is called intermittency in the context of chaos. In an open fluid the laminar and turbulent regions are divided by irregular and moving boundaries. The flow velocity sampled at a given point near such boundary will show alternating laminar or turbulent behaviour when the point enters or leaves the laminar region as the boundary moves. In one word, temporal intermittency is caused by spatially changing boundaries separating different flow regimes. However, in this volume we use the term intermittency exclusively to refer to random alternations of chaotic and regular behaviour in time evolution without involving any spatial degrees of freedom.

Among the various routes to chaos mentioned in Chapter 2, period-doubling and intermittency are in fact twin phenomena. Inspecting. the bifurcation diagram of a one-dimensional mapping or the "phase diagram" showing the bifurcation and chaos structure of a system of ordinary differential equations, e.g., Fig. 1 in Paper 17, one sees many chaotic regions contained in between pairs of type RL^{n-1} and RL^{n} periods. (For the meaning of these letters see Chapter 4,) If one enters a given chaotic region from the RL^{n-1} side, then the transition takes place via period-doubling bifurcations. If one approaches the same chaotic region from the RL^{n} boundary, the intermittent transition shows itself. Both period-doubling and intermittency are described, by the same renormalization group. equation (4-24) with different boundary conditions[2] and are well understood by now. Recommending the reader to consult the reprinted papers and reviews for details, we only touch briefly a few points related to the intermittent transition.

1. The mechanism of mtermittency in one-dimensional mappings has been fully understood. It occurs in the neighbourhood of a tangent bifurcation, whereas

period-doubling is related to pitchfork bifurcations. (For figures and explanation of these bifurcations see, May (1976), Paper 6 in this volume.) Take for example the logistic mapping (4-6). When the parameter μ is kept very close to, but still less than μ_c = 1.75, its third iterate $F(3, \mu, x)$ (we use definition (4-4) for iterates) has three humps or valleys that come very close to the bisector in the F versus x diagram, but does not touch it, leaving thereby very narrow corridors between the curve $F(3, \mu, x)$ and the bisector (see Fig. 4 in Paper 20). Every time when one iterate point falls near the entrance to one of these corridors, it will take many iterations to pass through the corridor. When μ is very close to μ_c, this looks much like a segment of convergent iteration to the would-be fixed point at μ_c and leads to the laminar phase of iteration. Having passed one corridor, the point makes a number of large-step jumps before falling again to the entry of one or another corridor. These jumps constitute the random bursts of the turbulent phase. The nearer μ is to μ_c, the longer the averaged laminar time. Simple mean-field arguments give the passage thne t^3

$$\tau \text{ a } (\mu_c - \mu)^{-(1-1/z)} \quad , \tag{6-1}$$

where z denotes the order of maximum in the mapping (4-5). For the usual case $z = 2$ we have $t \propto (\mu_c - \mu)^{-1/2}$ which is to be compared with the divergence rate of correlation length ξ near a continuous phase transition point T_c

$$\xi \alpha |T_c - T|^{-\nu} \tag{6-2}$$

with its mean-field value $\nu = \frac{1}{2}$.

2. Intermittent transitions had been observed in computer experiments on ODE's before its mechanism was explained by Pomeau and **Manneville[4]**. In periodically forced systems, mtermittency associated with tangent bifurcations of not very long period can be easily distinguished from chaotic or transient behaviour by their specific subharmonic. stroboscopic sampling diagrams[5]. Another way to explore the intermittent transitions consists in constructing the (P_i, P_{i+n}) map near a tangent bifurcation of period n, where P_i may be any of the variables under study. If there were an exact period n, one would get n points on the bisector of the (P_i, P_{i+n}) diagram, but at the intermittent transition we can only see the "spirit" of the would-be fixed points, i.e., n clusters along the bisector together with a few points scattered away from it. This is a very effective method to recognize intermittency in practice.

Intermittent transitions have been reported in many laboratory experiments, e.g., in chemical turbulence, thermoconvective instability and nonlinear electronic **circuits[6]**. We would like to point out that since there are infinitely many periodic windows born from tangent bifurcations embedded in the chaotic regions of the parameter space, the intermittent transitions are typical phenomena to be observed in numerical and laboratory experiments.

3. A few words about the renormalization group equation for intermittent transitions. We know from Chapter 4 that a periodic window In one-dimensional mapping remains stable whenever the absolute value of its derivative does not exceed one (see (4-11)). In fact, the limit F' = -1 correspondsto period-doubling bifurcations, whereas, F' = +1 indicates an intermittent transition. Therefore, it is not surprising that intermittent transitions can be described by the same renormalization group equation of Feigenbaum

$$g(x) = -\alpha g(g(-x/\alpha)) \tag{6-3}$$

with changed boundary conditionsa

$$g(0) = 0, \qquad g'(0) = 1 \quad , \tag{6-4}$$

(cf. (4-25)). However, it is remarkable that Hirsch, Nauenberg and **Scalapino[2]** found an exact solution to (6-3) and (6-4). Their solution has been extended to give all the intermittent exponents and universal functions g(x) for general z [7].

4. Our last remark concerns the effect of external noise on intermittency. Contrary to what one might expect by Intuition, a small amount of noise increases the averaged passage **time[8]**.

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Chapter 7
THE TRANSITION FROM QUASIPERIODICITY TO CHAOS

The transition from quasiperiodicity to chaos has become a hot subject since 1982. This is provoked by the interest to elucidate whether there are universal aspects in the Ruelle-Takens route to chaos, which remains less understood theoretically, in comparison with the period-doubhng and intermittent routes.

As mentioned in Chapter 3, Ruelle and **Takens**[1] proposed an alternative to the Landau-Hopf picture of infinitely increasing number of incommensurable frequencies for the onset mechanism of turbulence, They showed that quasiperiodic motion on a 4-torus, i.e., with 4 incommensurable frequencies, is in general unstable and can be perturbed into a strange attractor corresponding to turbulent motion. A few years later, in collaboration with Newhouse they sharpened the above result to that of a 3-torus, i.e., quasiperiodic motion with three incommensurable frequencies is in general unstable and can be perturbed into chaotic motion[2].

Let us denote quasiperiodic motion with n incommensurable frequencies by Q_n. The Landau-Hopf route to chaos can be expressed as $Q_n \xrightarrow{n\to\infty}$ **chaos.** Ruelle and Takens first suggested the possibility of $Q_3 \to$ chaos, and then estimated it as nongeneric and replaced it by $Q_2 \to$ chaos.

In Ref. 2 the perturbation required to destroy, Q_3 was rather specific, so in recent years the question has been raised concerning the existence of stable Q_3 in dynamical systems ad a generic, i.e. typical, possibility. Indeed, stable Q_3 has been shown to exist in several mathematical models and in hydrodynamical **experiments**[4]. Therefore, what was considered by Ruelle and Takens as nongeneric cases may well happen to be allowable in nature and there occurs the necessity to understand the mechanism of $Q_3 \to$ chaos transition.

Anyway, the transition $Q_2 \to$ chaos is generic and it does take place in many models and laboratory experiments. Two **groups**[5] studied the $Q_2 \to$ chaos transition by using renormalization group idea and discovered universal characteristics for this

transition. Since it is impossible to have quasiperiodic motion in purely one-dimensional mappings, both groups turned to the next simplest model – the mapping of a circle onto itself[6] :

$$\theta_{n+1} = \theta_n + \Omega - \frac{\mu}{2\pi} \sin 2\pi\theta_n \quad (\text{mod } 1) \tag{7-1}$$

which is a limiting case of the two-dimensional annular mapping[5].

In the linear case $\mu = 0$ the mapping

$$\theta_{n+1} = \theta_n + \Omega \quad (\text{mod } 1) \tag{7-2}$$

has only two regimes: periodic for Ω rational and quasiperiodic for Ω irrational. (It is sufficient to restrict Ω to the interval (0, 1) due to taking (mod 1).) Nonzero μ introduces nonlinearity into the system and causes the periodic motion to persist for wider range of Ω values. This is the simplest occurrence of frequency-locking in nonlinear oscillator systems (see Fig. 7-1).

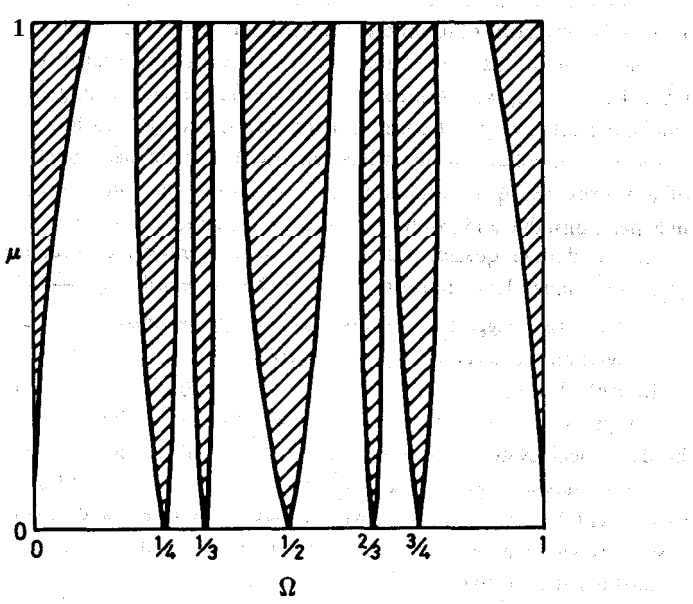


Fig. 7-1 Frequency-locking zones in the circle mapping (schematic).

The order of occurrence of the frequency-looking regions is given by the *Faray sequences* known in number theory[7] . An n-Faray sequence is the increasing succession of rational numbers p/q whose denominators are less than or equal to n, e.g., for n = 5 we have

0/1, 1/5, 1/4, 1/3, 2/5, 1/2, 3/5, 2/3, 3/4, 4/5, 1/1

(The case n = 4 is shown schematically in Fig. 7-1.)

Since for $\Omega = p/q$, p and q being relative prime, the mapping (7-2) has period q, it is the denominators in the Faray sequences which correspond to. the observed periods. For $\mu = 0$ it is clear that periodic regimes can in no way fill up the whole Ω interval, because rational numbers have measure zero on the real axis. With increasing μ the frequency-locking "tongues" in Fig. 7.1 get wider and wider. They, will contribute a positive. measure on Ω. This picture persists for all μ less than one. In other words, when $\mu < 1$ we have periodic regimes immersed in the sea of quasiperiodicity. Chaotic motion can occur only for $\mu > 1$. Even for μ approaching 1 from below the periodic tongues do not approach each other closely and quasiperiodic gaps are left where the $Q_2 \rightarrow$ chaos transitions can take place.

However, it is impossible to stay within the quasiperiodic regime all the time when the parameters are varied. Sooner or later frequency-locking is destined to get in. Since the more irrational the Ω value is, the longer the quasiperiodic regime persists, as we have learned from the destruction of invariant KAM curves in conservative systems (cf. Chapter 2), it was suggested in Ref. 5 to stick to a sequence of rational numbers p_i/q_i which converges to the "most irrational" number the Golden Mean $(\sqrt{5}-1)/2 = 0.61803\ldots$.

There are various ways to compare whether one irrational number is more irrational than another. Here the continuous fraction representation of numbers is used. Every irrational number can be represented by a unique infinite continuous fraction

$$(a_1, a_2, a_3, \ldots) \equiv \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{\ldots}}}} \tag{7-3}$$

If one asks how close a truncation of (7-4) approximates the irrational number it represents, then the worst fraction must be that with all a_i's equal to 1. We take this fact as the definition of the most irrational number, This number is nothing but the Golden Mean. It is known from elementary number theory that the truncations of the continuous fraction (1, 1, 1, 1, . . .) are given by the ratios of Fibonacci numbers:

$$\Omega_i = \frac{F_i}{F_{i+1}}, \qquad i = 1, 2, 3, \ldots \tag{7-4}$$

where $F_0 = 0$, $F_1 = 1$, and

$$F_{i+1} = F_i + F_{i-1} \quad . \tag{7-5}$$

Instead of following a curve of quasiperiodic regime in the (μ, Ω) plane, it was suggested in Ref. 5 to stick to the sequence (7-4) up to the llmit $i \to \infty$ where a transition from quasiperiodicity to chaos takes place. This can be done only when two parameters μ and Ω are both varied. In the language of dynamical system theory this is called a *codimension* two bifurcation sequence:

Using the recurrence relation (7-S) for Fibonacci numbers a pair of renormalization group equations similar to the Feigenbaum equation (4-24) was derived[5]:

$$f(x) = \alpha f(\alpha(f(\alpha^{-2} x)))$$
$$f(x) = \alpha^2 f(\alpha^{-1} f(\alpha^{-1} x)) \tag{7-6}$$

and numerical study was carried out. We mention in passing that these two equations are not independent[8].

Being unable to present all the results obtained so far[9] on the $Q_2 \to$ chaos transition in this short chapter, we point out only that the transition is accomplished via loss of differentiability of certain functions which are analytic for $\mu < 1$. If one goes back to the case of twodimensional annular mapping, the loss of differentiability can be visualized as progressively nested kinking of the invariant curve at all length scales.

Due to the presence of "rotational symmetry" in the annular mapping the destruction process takes place everywhere along the invariant curve which cannot be the general rule, say, in systems of ODE's. Indeed, there have appeared a few reports[10,11] on the $Q_2 \to$ chaos transitions in ODE's. A common feature of these numerical observations is the kinking and folding of the originally smooth Poincare map at one or more points, but not everywhere simultaneously. These transitions

are almost imperceptible at the beginning, but are reflected more clearly in the power spectra[11].

Besides the "local" universality studied in Ref. 5, there, seems to be also similarity in the global bifurcation structure leading to the $Q_2 \to$ chaos transition. At least in some idealized models[7,11] the transition should be elucidated in the light of the relation between the Farray sequence immersed in the quaslperiodic sea and the U-sequence (see Chapter 4) immersed in chaotic regimes. Thls question is far from being solved at present.

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Chapter 8

CHARACTERIZING THE ATTRACTORS: DIMENSION AND LYAPUNOV EXPONENTS

We mentioned in Chapter 5 that the strategy in studying the bifurcation and chaos "spectrum" of a physical system consists in first identifying various periodic regimes with confidence and then trying to characterize the chaotic motion on the attractors. Both in the laboratory and on computers it is very difficult to distinguish a purely chaotic motion from a quaslperiodic one or from a motion perturbed by external noise by simply looking at the finite sequence of data.. Moreover, although mathematicians have prepared such nice notions as hyperbolic attractor, Axiom A system or Smale's horseshoe, they are not directly applicable to systems of physical interest. Consequently, the term "strange attractor" is now being used in physical literature very liberally. Anyway, we need some quantitative means to recognize, characterize and classify attractors. It is also desirable to measure the strangeness of strange attractors from the data available, in particular, from a single time series of sampled points.

Power spectrum analysis is useful in telling quasiperiodic motion from periodic, and in identifying high order periodicities embedded in chaotic bands by the fine structures in spectra. Power spectra of chaotic attractors associated with period-doubling distinguish themselves by sharp peaks on the background of broad-band noise, but it makes little difference between attractors of other types. We indicate a few papers on scaling property of power spectra[1] and turn to more sophisticated ways of characterizing attractors, among which Lyapunov exponents and various definitions for dimension are widely used.

Lyapunov exponents, dimension, measure and entropy (in a mathematical sense) are closely related notions. They are backed up by a few mathematical theorems, complemented by plausible conjectures and numerical experience. We devote this chapter to a brief summary of the most important concepts and techniques.

The phase space volume of a dissipative system contracts in the process of evolution and the motion is confined to a certain attractor in the long time limit $t \to \infty$. The dimension of the attractor D is lower than the dimension n of the original phase space. (No Liouville's theorem holds for dissipative systems.) Using any reasonable definition of dimension, the dimension of trivial attractors can easily be calculated to be integers: $D = 0$ for fixed points, $D = 1$ for limit cycles, $D = 2$ for 2-torus, etc. However, the dimension of strange attractors often turns out to have noninteger value, yet what is an object with noninteger dimension?

Hausdorff introduced a generalized definition of dimension as early as in 1919, It can be. explained very simply in the following way. Take usual "regular" geometrical object, say, a cube, and double its linear size in each spatial directions. We get a cube whose! volume is eight times larger than the original one, because $2^3 = 8$. In general, taking an object of dimension D and increasing its linear size in each spatial directions ℓ times, one would have its volume increased to $k = \ell^D$ times of the original. Inverting this simple relation, we get a new definition for dimension:

$$D = \frac{\ln k}{\ln \ell} \tag{8-1}$$

Now, we have freed ourselves from the restriction of D being an integer.

A precise definition of ***Hausdorff dimension*** requires the notion of Hausdorff measure (see the books by **Mandelbrot[2]**), but formula (8.1) is enough for our practical needs. We begin with the simplest example of a geometrical object having noninteger dimension — the Cantor set.

Take the interval (0, 1) and delete the central one third (1/3,2/3), then repeat this operation with respect to the remaining segments again and yet again, ***ad infinitum.*** What is the dimension of the limiting set of points thus obtained? Let points left in the interval (0, 1/3) constitute our geometrical object under consideration. Increasing. its linear size by a factor $\ell = 3$ would yield two copies ($k = 2$) of the same object, therefore

$$D = \frac{\ln 2}{\ln 3} = 0.6309\ldots \tag{8-2}$$

Actually, many strange attractors have a Cantor set like structure (see, e.g., the frequently cited Figs. 3-6 in Henon (1976), Paper 13 in this volume). To calculate the dimension numerically, a box-counting algorithm may be used. One divides that part of the phase space where the attractor lies into small cells of linear size ϵ, and counts the number $N(\epsilon)$ of such cells that contain at least one point

of the orbit. Then, the dimension is calculated by taking the limit

$$D_C = \lim_{\epsilon \to 0} \frac{\ln N(\epsilon)}{\ln(1/\epsilon)} \tag{8-3}$$

To be precise, (8-3) is called the *Kolmogorov capacity* of the limiting set of points, hence the subscript C to D_C. In practice, one counts $N(\epsilon)$ for a series of finite ε's and obtains D_C from the slope of the $\ln N(\epsilon)$ versus $\ln(1/\epsilon)$ plot.

Thus far in counting N(ε) no attention has been paid to the possible inhomogeneity of the attractor, i.e., no matter how many times the orbit travels through a given cell, the cell is counted only once. To correct this inexactitude one introduces a weight according to how frequently a cell is visited. If the i-th cell is visited with a probability P_i, then the definition (8-3) is replaced by

$$D_I = \lim_{\epsilon \to 0} \frac{I(\epsilon)}{\ln(1/\epsilon)} \tag{8-4}$$

where

$$I(\varepsilon) = -\sum_{i=1}^{N(\epsilon)} P_i \ln P_i \quad , \tag{8-5}$$

N(ε) being the total number of cells visited. If every cell is visited with equal probability, i.e., P_i = l/N(ε), then (8-5) reduces to $I(\varepsilon) = \ln N(\epsilon)$ and $D_I = D_C$. In general $D_I \leq D_C$. D_I is called the *information dimension.*

There exists a number of dlfferent ways to define **dimension**[3]; see especially the review by Farmer et al. (1983). It is, conjectured that the Hausdorff dimension is equal to D_C and some other defintions are equivalent to D_I. Numericalcalculations seem to support this conjecture.

The box-counting algorithm to compute the dimension is very time-consuming and becomes impractical when the dimension of the original phase space n ≥ 2. Moreover, the dimension reflects only one essential aspect of dissipative dynamics, namely, the contraction of phase volume. However, a strange attractor cannot be generated merely by contraction. It must be stretched and folded in some directions as well. To describe these more subtle features we need the notion of *Lyapunov exponent* and we shall see that the dimension can be readily calculated from these exponents. In addition, Lyapunov exponents furnish' us with a classification scheme of attractors.

Consider an autonomous systems of ODE's

$$\dot{x}_i = f_i(x_1, x_2, \ldots, x_n) \quad , \qquad i = 1, 2, \ldots, n \quad . \tag{8-6}$$

It can be linearized near a given point $\mathbf{x}_0$

$$x_i = x_{0i} + \delta x_i \qquad i = 1, 2, \ldots, n$$

to yield a linear, but nonautonomous (depending on $\mathbf{x}_0$) system of equations

$$\frac{d}{dt}\delta x_i = \sum_j^n U_{ij}(x_0)\delta x_j \qquad i = 1, 2, \ldots, n \tag{8-7}$$

where

$$U_{ij}(x_0) = \left.\frac{\partial f_i}{\partial x_j}\right|_{\mathbf{x}=\mathbf{x}_0} \quad . \tag{8-8}$$

The eigenvalues of matrix (8-8) determine the local stability of Eq. (86) in the vacinity of $\mathbf{x}_0$: if the real parts of some eigenvalues happen to be positive numbers, then nearby trajectories wlll run away at an exponential rate. In principle, one can average the real parts of all eigenvalues to get a set of global characteristics — the Lyapunov exponents. However, there is another more intuitive way to calculate n real numbers playing the same role[4].

Let us build a local coordinate system with origin at $\mathbf{x}_0$ by choosing at random n small vectors coming out from $\mathbf{x}_0$ and then orthogonalizing them. Denote these vectors by $\{\mathbf{e}_0^{(1)}, \mathbf{e}_0^{(2)}, \ldots, \mathbf{e}_0^{(n)}\}$. Then integrate the Eq. (8-6) one time step forwards to reach a new position $\mathbf{x}_1$, and at the same time let $\{\mathbf{e}_0^{(i)}\}$ evolve according to the linearized Eq. (8-7) to form n new vectors with origin at $\mathbf{x}_1$. Now we can calculate a number of goemetrical ratios :

C_1^n = n length ratios $|\mathbf{e}_1^{(i)}| \,/\, |\mathbf{e}_0^{(i)}|$;

C_2^n = $n(n-1)/2$ parallelogram area ratios $|\mathbf{e}_1^{(i)} \wedge \mathbf{e}_1^{(j)}| \,/\, |\mathbf{e}_0^{(i)} \wedge \mathbf{e}_0^{(j)}|$;

. . . . . .

C_n^n = 1 n-dimensional hyperparallelepiped volume ratio

$$\frac{|\mathbf{e}_1^{(1)} \wedge \mathbf{e}_1^{(2)} \wedge \ldots \wedge \mathbf{e}_1^{(n)}|}{|\mathbf{e}_0^{(1)} \wedge \mathbf{e}_0^{(2)} \wedge \ldots \wedge \mathbf{e}_0^{(n)}|} \quad ;$$

(The exterior product ∧ generalizes the vector product x to higher dimensional spaces:) Being geometrical quantities, all these ratios are real numbers. Repeating

this process step by step along the trajectory and taking the long time average of these numbers, we get a set of global characteristics of the attractor. It was proved by Benettin et al. (1976) that provided the vectors $\{ \mathbf{e}_0^{(i)} \}$ are taken at random, the length ratios converge to the maximal Lyapunov exponent λ_1, the parallelogram area ratios converge to the sum of the two largest Lyapunov exponents $\lambda_1 + \lambda_2$, . . . the n-dimensional hyperparallelepiped volume ratio converges to the sum of all Lyapunov exponents $\lambda_1 + \lambda_2 + \ldots + \lambda_n$. The last quantity must be equal to the contraction rate of a volume element along the trajectory, i.e.,

$$\lambda_1 + \lambda_2 + \ldots + \lambda_n = - \operatorname{div} \mathbf{f} \tag{8-9}$$

where $\mathbf{f} = (f_1, f_2, \ldots, f_n)$ is the vector field on the right hand side of Eq. (8-6). In practice, Eq. (8-9) serves as a numerical check for the correctness of computation. In stating the results above, we have introduced an ordering for all Lyapunov exponents:

$$\lambda_1 > \lambda_2 \geq \ldots \geq \lambda_n \tag{8-10}$$

We mention in passing that the step by step orthogonalization appears to be crucial in numerical calculation, since the initially independent vectors tend to evolve into a subspace of lower dimensionality due to phase volume contraction. For details see Shimada and Nagashima (1979), Paper 27 in this volume.

Kaplan and Yorke (1979) conjectured that the Kohnogorov capacity D_C is related to the Lyapunov exponents by

$$D_C = j + \left(\sum_{i=1}^{j} \lambda_i \right) \Big/ |\lambda_{j+1}| \tag{8-11}$$

where j is the number of Lyapunov exponents which assures the non-negativeness of the sum in (8-11), i.e.

$$\sum_{i=1}^{j} \lambda_i \geq 0.$$

In the extreme cases $j = 0$ or n, define $D_C = 0$ or n respectively.

Mori (1980) "derived" another relation between dimension and Lyapunov exponents

$$D_C = m_0 + m_+ (1 - \lambda_+ / \lambda_-) \quad , \tag{8-12}$$

where

$$\lambda_+ = \frac{1}{m_+} \sum_{\lambda_i > 0} \lambda_i \quad , \quad \lambda_- = \frac{1}{m_-} \sum_{\lambda_i < 0} \lambda_i \quad ,$$

and m_-, m_0, m_+ denote the number of negative, zero, and positive exponents respectively. For sufficiently large n (8-11) differs from (8-12) significantly, because all negative exponents contribute to (8-12), whereas only a finite number of them enters (8-11). Numerical experiments seem to be in favour of (8-11). In general, the Lyapunov exponents converge fairly rapidly and provide an effective way to calculate the dimension D_C in comparison to the box-counting algorithm.

Yet the largest Lyapunov exponent alone is of much help to numerical work, since the appearance of a positive exponent signals the onset of chaos. It was suggested by many authors to put forward'a qualitative classification of attractors according to the signature of the Lyapunov exponents. For example (-, -, -) corresponds to fixed point with $D = 0$, (0, -, -) to limit cycle with $D = 1$, (0, 0, -) to 2-tori with $D = 2$, and (+, 0, -) to strange attractor with $D = D_C$. In this scheme a signature (+, +, 0, -) would correspond to strange attractor displaying *superchaos*. Since the signature changes with control parameters, they provide also a means to classify transitions between various regimes.

Dimension and Lyapunov exponent for one-dimensional mappings have some peculiarities worth mentioning. First, there is only one Lyapunov exponent, sometimes called also Lyapunov number

$$\lambda(\mu) = \lim_{k \to \infty} \frac{1}{k} \sum_{i=0}^{k-1} \ln |f'(\mu, x_i)| \quad , \tag{8-13}$$

Periodic orbits correspond to $\lambda < 0$, chaotic orbits to $\lambda > 0$.

Second, $\lambda = 0$ at every bifurcation point, at the accumulation point μ_∞ of the period-doubling cascade, and at tangent bifurcations. Grassberger (1981) showed that at μ_∞, where the attractor has a Contor set structure, the Hausdorff dimension is universal, with value $D_C = 0.538\ldots$ On the other hand, due to critical slowing down one always gets a cluster of points near every finite bifurcation point μ_k, no matter how long the iteration is being carried on. Wang and Chen[6] proved that this cluster acquires an "operational" dimension $D = 2/3$, if measured by a box-counting algorithm, although the true dimension of a periodic orbit must be zero. There exists a scaling function relating these two values $D = 2/3$ and $D = 0.538\ldots$[6].

Another remark concerns the reconstruction of strange attractors and the measurement of their dimension from experimental data[7]. For a real. physical system neither the dimension of the phase space, nor that of the attractor are known in advance, therefore it has been suggested to extract information on

dimensional&y from single time series of the experimental data by linear regression and other methods of statistical estimation, For the time being only the integer part of the dimension can be evaluated. A few "real" strange attractors have been reconstructed in this way, e.g., the dimension of strange attractor in turbulent Couette flow was estimated to be 5 for not very high Reynolds number (Brandtiter) et al. 1983).

We conclude this chapter by discussing briefly the relation of Lyapunov exponents to measure and entropy defined on strange attractors*. Chaotic motion on strange attractors can be characterized by their ergodic properties. In particular, certain continuous measure exists on the attractor and a metric entropy (the K-entropy) invariant under measure-preserving transformations may be defined. K-entropy is essentially the sum of all positive Lyapunov exponents. Therefore, it is closely related to the Hausdorff dimension. Being metric properties, dimension, K-entropy, and Lyapunov exponents may change under continuous transformation of the phase space, e.g., when going from one coordinate system to certain other systems. In this context they are not as simple as topological characteristics such as the topological dimension or the topological entropy. However, it is just this flexibility which makes these metric properties more relevant to physics. In view of their possibly increasing role in physics we give some references to this more or less mathematical topic [8].

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Chapter 9
SCALING FOR EXTERNAL NOISE

We have emphasized the intrinsic nature of chaos in Chapter 1 and throughout the chapters thus far. However, external noise is an unavoidable factor in computer experiments (round-off errors) and laboratory experiments (thermal fluctuations of the ambience). A fuller understanding of chaos requires the inclusion of external noise into the theoretical framework.

Actually, in the theory of chaotic phenomena external noise plays a more constructive role than just to be an undesirable but inescapable disturbance from the outside. Only with the external noise taken into account shall we have a full parallel with the theory of continuous phase transitions which has enriched physics with so many new concepts since the mid-60's.

Phase transition occurs at certain critical value of control parameter such as the temperature T or the pressure p when the latter is subjected to a slow and progressive variation. Many thermodynamical quantities show singular behavior when $|T - T_c| \to 0$, where T_c denotes the critical point. Chaotic transition takes place at critical value μ_c of the control parameter such as the Reynolds number and shows up as singularity in $|\mu - \mu_c|$. The formation of the new phase may be described by the appearance of nonzero "order parameter", say, macroscopic magnetization in ferromagnetic phase transitions. Chaotic state can be characterized by certain "disorder parameter" as well. It was thought by some authors that the Lyapunov exponent may be taken as disorder parameter[1]. In fact, the Lyapunov exponent corresponds to the reciprocal of correlation length ξ in phase transitions as we shall see soon. One must take some other statistical characteristic of the chaotic motion on the strange attractor (entropy or the like) to be the disorder parameter.

Moreover, a new ordered phase may appear either spontaneously at the critical point T_c, or may be induced by suitable external fields. For example, the external

magnetic field plays the role of "ordering field" in magnetic phase transitions. In phase transition theory the ordering field coupled to the order parameter is sometimes called a dual field. Doesthere exist a "disordering field" for chaotic transition? Chaotic states are characterized by spontaneous appearance of seemingly random motion showing noisy broad bands in the power spectra. Therefore, external noise may well play the role of the disordering field. To include the external noise, one adds a noisy source term to the nonlinear mapping (4-2) and thereby changes it into a discrete Langevin equation:

$$x_{n+1} = f(\mu, x_n) + \sigma\xi_n \quad , \tag{9-1}$$

where ξ_n are random numbers obeying certain statistical distribution with the constraints

$$\overline{\xi_n} = 0, \quad \overline{\xi_n \xi_m} = \delta_{nm}$$

and the coefficient σ, being a measure of the noise strength, can be compared with the external magnetic field in magnetic phase transitions.

An important notion in phase transition theory is the correlation length ξ which diverges at the critical point as

$$\xi \propto |T - T_c|^{-\nu} \tag{9-2}$$

where ν is a positive universal exponent. So far in this volume, chaotic behaviour has been considered only in connection with time evolution of nonlinear systems (except for a few words in Chapter 4 on PDE's). From the discussion in the last chapter we know that when the Lyapunov exponent λ is negative, the influence of initial conditions decays quickly and the system approaches an asymptotic state independent of the initial conditions, whereas when λ is positive, nearby orbits will run away exponentially and any minor change in initial conditions will be amplified with time. Therefore, λ^{-1} can be taken as a measure of time correlation. Since λ passes through zero at bifurcation points, in particular, at the limiting point μ_∞ of period-doubling sequence, one expects a relation similar to (9-2) to hold :

$$\lambda^{-1} \propto |\mu - \mu_\infty|^{-t} \tag{9-3}$$

Starting from the discrete Langevin equation (9-l) the whole arsenal of critical dynamics in phasetransition theory (see, e.g., Ref. 2 for a review) can be brought to bear on chaotic transitions. We have included three pioneering papers along this

line into this volume[3-5] (see also papers[6] on Champman-Kohnogorov equation related to the discrete Langevin equation). The main result of Ref. 5 was the scaling relation for λ in the presence of external noise:

$$\lambda(\mu_\infty - \mu, \sigma) = (\mu_\infty - \mu)^t \Phi\left(\frac{\sigma^\theta}{(\mu_\infty - \mu)^t}\right) \tag{9-4}$$

where Φ is a universal function and

$$t = \frac{ln2}{\ln \sigma} = 0.4498069\ldots$$

$$\theta = \frac{\ln 2}{\ln k} = 0.366739\ldots \tag{9-5}$$

k = 6.6190. . . being a new exponent related to the scaling of external noise. Putting $\sigma = 0$ in (9-4), we are led to

$$\lambda(\mu_\infty - \mu, 0) \propto (\mu_\infty - \mu)^t \tag{9-6}$$

which has the desired form of Eq. (9-3) and was obtained without involving external noise',

The scaling relation derived in Ref. 4 looks slightly different:

$$\lambda(\mu_\infty - \mu, \sigma) = \sigma^\theta L\left(\frac{\mu_\infty - \mu}{\sigma^{\theta/t}}\right) \tag{9-7}$$

It suffices to redefine the scaling function by

$$xL(1/x) \equiv \Phi(x) \tag{9-8}$$

with $x = \sigma^\theta/(\mu_\infty - \mu)^t$, to see that Eq. (9 -7) is identical to (9-4).

Now we are prepared to summarize the role of external noise in chaotic dynamics as follows. First, external noise will wash out the details in high order bifurcation sequences including both the period-doubling cascades and the inverse sequences of chaotic bands[3]. In order to see one more bifurcation one must decrease the noise level by a factor of k. This explains the meaning of the noise exponent k. Second, the presence of external noise makes the bifurcation diagram fuzzy and lowers the accumulation point μ_∞ slightly as it can be expected from the role of noise as disordering field. Third, the Lyapunov exponent passes through zero at various bifurcation points, but this is no longer true when external 'noise is added. For more details see review papers[8].

The role of external noise was studied mainly on the logistic mapping[9]. There were a few reports on the effect of noise on strange attractors[10]. The influence of noise on the $Q_2 \rightarrow$ chaos transition and the intermittent transition has been studied in Ref. 11 and Ref. 12 respectively.

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Chapter 10
EXPERIMENTAL OBSERVATION OF CHAOTIC PHENOMENA

Chaotic phenomena are not just a motley collection from the mathematical labyrinth, but a wide class of natural events found in the physical world. Generally speaking, chaos happens more frequently than order, just as there are many more irrational numbers than the rationals. Chaos can be thought of as a new regime of nonlinear oscillations, as a compromise. between competing periodicities, as overlap of resonances, as accumulation of many instabilities, as the prelude to turbulence. Therefore, it is not difficult to imagine various experimental situations generating chaotic behaviour.

Moreover, Nature is always richer than mathematical models. Experimental study of chaos has at least a twofold mission: to verify the theoretical understanding gained from model studies and to bring about new physics by challenging the existing theory with unexpected findings, not to mention the prospect for technological development at our present level of knowledge. From this view-point laboratory observations can be roughly divided into three groups: those almost coinciding with theoretical expectations, e.g., experiments using nonlinear circuits; those showing much more variations and complications, only some facets of which resemble one or another aspect of theoretical models, e.g., experiments on hydro0 dynamical instabilities; and those occupying an intermediate state, e.g., most of the experiments on acoustic, optical, chemical, and solid-state turbulence.

In what follows various kinds of experiments will be listed briefly along with references to the bibliography, including some theoretical analysis and proposals which come under the same heading. Whenever appropriate we shall make a few remarks on the experimental results.

1. Forced vibmtion of shallow water wave in a finite container[1]

Historically this was the first experiment where transition to chaos would have been discovered, if Faraday had the idea of period-doubling route to chaos (see Faraday in Ref. 1). In fact, Faraday observed subharmonic component $f_0/2$ in the shallow water wave in a container forced to vibrate vertically at frequency f_0. Lord Rayleigh noticed, repeated and wrote about this experiment in his *Theory of Sound,* § 68b. Modern data acquisition techniques helped to reveal much longer bifurcation sequences and the onset of chaos[1]. However, the authors of Ref. 1 described their results as departures from period-doubling. For example, taking the driving period to be 1, they observed a sequence

1, 2, 4, 12, 14, 16, 18, 20, 22, 24, 28, 35,

but the power spectrum given for period 14 showed clearly a 2*7 fine structure, testifying to its being a secondary period 7 embedded in a chaotic band of period 2. Since the bifurcation and chaos "phase diagram" of, a real physical system may be very intricate due to bending, folding and interpenetration of various sequences of periods (cf. Fig. 1 in Paper 17, or Fig. 1 in Paper 18 in this volume), it is very easy to mix up periods, coming from different sequences or from different parts of the same sequence.

2. Bifurcation and chaos in nonlinear circuits

The most complete and beautiful results on chaos have been obtained on nonlinear circuits, because the experimental conditions can be precisely controlled and the circuits can be well represented by ODE's with only a few variables.

Earlier work on nonlinear circuits[2] were carried out on systems of many coupled nonlinear oscillators. Complicated combinations of periods and chaos were indeed observed, but could not be fitted into simple systematics as has now been done.

Most of recent experiments[3] deal with a single nonlinear oscillator driven by a periodical signal. Period-doubling bifurcations, tangent bifurcations, intermittency, crisis of the attractors, and the effect of external noise have been studied. In particular, the Feigenbaum constants δ and α, the noise scaling exponent k, and the ratio of successive period-doubling peaks in the power spectra were measured to be in good accord with that of the onedhnensional unimodal mappings. It seems desirable to carry out a detailed study on transitions from quasiperiodicity to chaos using nonlinear circuits, but probably more than one nonlinear oscillators should be used for this purpose.

3. Bifurcation and chaos in mechanical or electromechanical systems[4]

This is another group of experiments where results in keeping with theory can be

expected. Some of these systems may serve as classroom demonstration of chaotic phenomena. We add also that both the Lorenz and Duffmg equations can be modelled by simple mechanical systems (see book B32 and B36 respectively).

4. Onset of hydrodynamical instabilities[5,6]

Low temperature technique, laser Doppler velocimetry, and modern data acquisition systems have brought the study of turbulence back to physical laboratories. However, at the present time most of these studies still concentrate on instabilities in finite containers where the decay of many irrelevant modes of motion makes the results closer to theoretical predictions. As regards the more important shear flow instabilities, little progress has really been achieved with the help of the philosophy of chaos, because infinite degrees of freedom must be involved to interpret the experiments.

Thermoconvective instability of fluid in a container heated from below with free surface (the Benard instability) or without space in between the fluid and the upper plate of the container (the Rayleigh-Benard instability) has now become a classical setup to study the onset mechanism ofturbulence. Ahlersfirstcoriducted the experiment at low temperatures on liquid helium in 1974. Libchaber and Maurer (1980) observed the successive appearance of quasiperiodic regime with two incommensurable frequencies, frequency locking, and period-doubling. They resolved 2^n (n = 0 to 4) periods and chaotic bands with periods 2^n (n = 1 to 4) from the power spectra at small Prandtl numbers. Three-dimensional instabilities and intermittency were observed at larger Prandtl numbers. Rayleigh-Benard experiment using Hg as working substance revealed transition from quasiperiodicity with two or three incommensurable frequencies to chaos (Fauve and Libchaber, 1981).

Instabilities of Couette flow between rotating cylinders[6] show much more variations, some but not all of which may be put Into the framework of the Ruelle-Takens route. Especially when the outer cylinder also rotates or stops suddenly, many more very complicated patterns have been recorded. Most of these experimental findings still await a better theoretical understanding: The same may be said as regards experiments using rotating spheres (Belyaev et al., 1979).

5. Acoustic turbulence [7]

Subharmonic components have been known to exist in the spectra of cavitation (i.e., bubble formation in liquid subjected to high power supersonic Irradiation) noise. In searching for the period-doubling route to chaos, these experiments have been improved to get 2^3 subharmonics, but higher bifurcations cannot be identified due to the high level of external noise in this kind of devices.

Subharmonic generation has been observed also in supersonic absorption, in liquid helium. Above the superfluid transition temperature T_λ = 2.17 K, it does not differ from ordinary liquids. However, it is a little surprising that clear period-doubling bifurcation sequence and tangent bifurcation into periods 3, 5 and 7 show up at $T < T_\lambda$ when ordinary bubbles cannot form via the cavitation mechanism. Indeed, the first bifurcation occurs at the threshold of quantum vertex generation. Therefore, period-doubling in superfluid helium has a quantum origin, but enjoys the universality of other classical bifurcation phenomena. This is similar to universality in continuous phase transitions.

Other experiments on acoustic turbulence include simple acoustic systems in air, liquid nitrogen, or water.

6. Optical turbulence[8-10]

Lasers provide a typical class of nonlinear oscillator systems. The Maxwell-Bloch equations describing a unimodal laser in the semiclassical approach reduce to a system of three ODE's. Haken (1975) showed that this system can be transformed into the Lorenz model (5-4) under certain conditions, not realistic for a laser, but relevant to random spiking in the superradiance regime.

Ikeda (1979) predicted theoretically that bifurcation and chaos may appear in an optical bistability device using a ring cavity. This effect was first observed in so-called hybrid bistability devices by Gibbs et al. (1981), Paper 40 in this volume. Hybrid means that part of the optical output is delayed by an electronic circuit (or kept for a while by a microprocessor) and then fed back to the input signal. Recently, all-optical bistability devices have also been designed to show optical chaos, see, e.g., Harrison et al. (1983). The upper branch of the bistable states losses stability at certain threshold of the input power and gives rise to doubled periods $2t_R$, $4t_R$, and chaos, where t_R is the delay time. This kind of devices with feedback corresponds to time-delayed ODE's, which in the simplest case may be reduced to a single equation for the phase:

$$\tau\dot{\varphi}(t) = -\varphi(t) + f(\mu, \varphi(t - t_R)) \quad , \tag{10-1}$$

where τ is the intrinsic relaxation time of the nonlinear medium causing bistability, and $f(\mu, \varphi)$ is a nonlinear function of φ characterizing the cavity with nonlinear medium, μ and t_R being adjustable parameters. In the limit of long delay $\tau \ll t_R$ the derivative term in (10-l) may be neglected and the equation reduces to a one-dimensional mapping by setting $\varphi_n \equiv \varphi(n t_R)$:

$$\varphi_n = f(\mu, \varphi_{n-1}) \tag{10-2}$$

Now there is nothing strange to have period-doubling and chaos in such systems. However, in bistability devices chaos may appear in other regimes too, e.g., when $\tau \gg t_R$ (Ikeda et al., 1982).

In hybrid bistability devices usually the delay time t_R is of the order of milliseconds, so the frequency structure of the bifurcation and chaos has nothing to do with optics. Great effort has been made to search for purely optical chaos. Now we have quite a few reports on this subject[9], e.g., in Q-switched gas laser (Arechi et al., 1982). We mention also some general or theoretical discussion on optical turbulence",

7. *Chemical turbulence*

The concentrations of intermediate products in some specially designed chemical reactions oscillate with time. The most thoroughly studied reaction of this type, the Belousov-Zhabotinskii reaction, consists of some 20 elementary reactions. During the last decade these systems served as prototype for self-organization phenomena, i.e., spontaneous occurrence of temporal and/or spatial structures in the system when energy and matter flows are supported from the outside. 'If well stirred, the reacting system may be considered as spatially homogeneous and described by autonomous nonlinear ODE's, which often turn out to be isomorphic to some coupled-oscillator systems".

Reports on experimental observation of chemical turbulence began to emerge since 1977[12] and chaotic solutions have been found in mathematical models describing such systems".

In order to keep the reaction in steady state, some reactants and products must be supplied or removed from the system constantly. The flow velocity of these chemicals can be taken as control parameter and the concentration of certain intermediate products, say, the concentration of Br^- ions in the BZ reaction, is to be monitored. At very small flow velocity the system approaches thermal equilibrium without showing complicated time behaviour; At very large flow velocity the chemicals rush through the container, having no time to get into reaction. Therefore, only at intermediate flow velocities does one observe complicated dynamics, including multisteady states, simple or relaxation oscillations, intermittency, period-doubling and chaos, hysteresis, etc. It has been proved experimentally that such dynamical behaviour was surely caused by chemical reactions, but not by the hydrodynamics of the flow.

It was thought at the beginning that chemical turbulence offers something quite different from other nonlinear systems. However, the strange attractors reconstructed from experimental data happened to have sufficient low dimensionality

and the observed order of oscillation periods tits that of the U-sequence (q.v. Chapter 4). Therefore, people now tend to believe that in chemical turbulence there are more features in common, with other nonlinear systems. Moreover, apart from nonlinear circuits, chemical systems are easy to controland the dynamics involves a quite different frequency range compared to that of nonlinear circuits. All these make chemical turbulence an interesting subject deserving more attention.

8. *Chaotic phenomena in solid stare physics*[14-17]

Many models used in solid state physics consist of coupled or driven nonlinear oscillators whose time evolution must show chaotic behaviour within a suitable parameter range. On the other hand, competition of various commensurable or incommensurable periods makes chaos an intrinsic source for spatial disorder (or to be more precise, for order without periodicity). From numerous theoretical models and experimental proposals we mention only a few.,

It has been known since 1977 that in rf driven Josephson junctions used as parametric amplifier noise grows anomalously with the gain level. The equivalent noise temperature reaches 50 000 K whereas the device is kept at 4 K. Since such a high noise level cannot be explained by any known noise source and its amplification, Huberman et al. (1980) attributed it to the intrinsic dynamics of the junction. In the resistively shunted junction model the current equation reads

$$C\dot{V} + V/R + I_c \sin\varphi = I_r \cos\omega t + I_0 \tag{10.3}$$

where R, C, I_c, I_r, and I_0 denote the resistance, capacity, critical current, rf current and dc bias respectively. The phase difference across the junction is given by the Josephson equation

$$\dot{\varphi} = 2eV/\hbar \ . \tag{10-4}$$

In terms of dimensionless quantities these two equations lead to the following equation of a forced damped pendulum

$$\ddot{\varphi} + a\dot{\varphi} + \sin\varphi = i_0 + i_r \cos\Omega t \tag{10-5}$$

Both numerical study of (10-S) and experimental work revealed the appearance of chaos[14].

Another example of anomalous noise generation concerns the oscillatory circuit based on the Gunn diode. Nakamura[15] truncated the PDE describing the system into 40 coupled ODE's and found many attractors. He explained the anomalous

fluctuation by the interweaving basins of attraction. The equation of motion for dislocation lines in the presence of a supersonic field reduces to that of a periodically forced anharmonic oscillator, i.e., Eq. (5-8). Consequently, Herring and Huberman suggested to observe chaos in this **situation**[16].

Now let us turn to spatial chaos as a source of disorder. In solid state physics there are many cases of competing periods leading to the formation of new periodic or disordered structures and phase transitions may take place in between these structures. For example, a monolayer of inert gas atoms, adsorbed to a graphite substrate may form a two-dimensional lattice which does not necessarily match the lattice of the substrate. If the interaction between the monolayer and the substrate is weak enough, the adsorbed layer keeps its own periodicity. Conversely, when they interact strongly, the monolayer lattice is forced to match that of the substrate. In between these two extremes phase locking and spatial chaos may occur. This process resembles the behaviour of forced nonlinear oscillators with time evolution replaced by spatial ordering.

The simplest model for modulated structure was proposed by Frenckel and Kontorova in the 30's Suppose a one-dimensional chain of atoms is put in a cosine potential of the substrate, the n-th atom being at x_n. The Hamilton function of the system may be written as

$$H = \sum_n \frac{1}{2}(x_{n+1} - x_n - a)^2 + \frac{\mu}{(2\pi)^2}\cos 2\pi x_n \qquad (10\text{-}6)$$

where the period of the substrate field has taken to be 1, and μ measures the coupling. The equilibrium positions of atoms are determined by solving $\frac{\partial H}{\partial x_n} = 0$, namely,

$$x_n - x_{n-1} - (x_{n+1} - x_n) - \frac{\mu}{2\pi}\sin 2\pi x_n = 0 \qquad (10\text{-}7)$$

Let $y_n = x_n - x_{n-1}$, we have

$$y_{n+1} = y_n - \frac{\mu}{2\pi}\sin 2\pi x_n$$
$$x_{n+1} = x_n + y_{n+1} \qquad (10\text{-}8)$$

This is precisely the "standard mapping" (5-2) listed in Chapter 5. Now it is natural to have periodic or chaotic solutions depending on the parameter μ. When the characteristic wave number of the modulated structure varies continuously, commensurable phases may occur repeatedly at rational values, leading to the so-called devil's staircase. A shnilar situation appears in some statistical models and in spin glasses. Ruelle went even further to pose the question whether there exist turbulent crystals. We recommend the reader to consult the reviews by Bak, and by Fisher and **Huse**[17].

9. Chaos in biology[18,19] and in ecology[20]

We digress a little from physics to mention the- impressive experiment with periodically stimulated beating of cultured chicken cardiac cells's, displaying phase-locking, period-doubling; and chaos. It was "modelled by a periodically forced oscillator and an attempt was made to connect the chaotic dynamics with cardiac arrhytmias.

There are "many' speculations on the role of chaos in **biology**[19]. Chaotic behaviour should exist in biochemical reactions. However, it is certainly too early to draw any far-reaching conclusion, at least in as much as chaos is a purely classical phenomenon.

In a sense the whole modem story of chaos has its origin in ecological models. Therefore, it is appropriate to conclude thls chapter by listing a few more references"'.

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